# The `ilmarinen` Package: A Practical Implementation Report

## Numerical Realization of Variational Neural Architecture Meta-Optimization

Companion implementation notes

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**Abstract**

This report is the practical companion to the analytical reference notes on variational and field-theoretic neural architecture meta-optimization. Where the analytical document develops the theory—least-complexity width as an atomic-measure problem, priced and geometric depth, depth-renormalization criticality [19, 20], and the reading of training as a least-action/path-integral principle—this document records what was actually *built*, how each analytical formulation was mapped (exactly, approximately, or not at all) to a numerical implementation, and how the resulting system behaves on synthetic and real data. We describe the six schema modules that realize differentiable architecture selection over a vocabulary of primitive operations, culminating in a *unified* sequence schema in which all nine implemented primitive realizations compete under a single architecture parameter. We report selection outcomes and accuracies—measured freshly from the current package, not reconstructed—on three synthetic tasks (copy, adding, associative recall) and five real UCR/UEA (the standard univariate and multivariate time-series archives) datasets, alongside matched-complexity fixed-architecture baselines, and we document the concrete network each metaoptimizer selects (primitive type, layer count, and parameter count). We then lift the fixed-size restriction: by pricing capacity and folding a differentiable complexity penalty into the architecture objective, width and depth become *decisions* of the metaoptimality criterion, and we report the resulting fit–complexity frontier and compact-and-accurate models. Upstream of architecture search we add a *representation front-end* that discovers, from the data's mutual-information structure, whether it is naturally a one-dimensional sequence, a two-dimensional grid, or unstructured—recovering exact image shapes (CIFAR-10 $32 \times 32$, Fashion-MNIST $28 \times 28$) via a stride-anomaly signal and routing each dataset to the matching schema; automatic tensorization improves both accuracy and architectural appropriateness where the data has non-sequence structure. Beyond architecture search, we document a *symmetry-discovery front-end*—a physicist's preprocessing stage that discovers the continuous and discrete symmetries of the data (via a Lie-derivative nullspace and equivariance testing respectively), quotients them out to reduce dimensionality, and only then hands a minimal representation to the metaoptimizer; we connect this reduction to the information bottleneck [25, 24], validate it on QM7 molecular data (where it cuts rotated-test error substantially), and are candid about its false-positive failure mode on smooth ordered data and the guards required to contain it. We are explicit throughout about the boundary between what the theory promises and what the implementation delivers: several of the sharpest analytical structures are realized only in approximate or heuristic form, and a number of the theory's negative predictions were confirmed empirically. A closing section collects open questions and future directions distilled from the development history.

# Contents

# 1 Purpose and scope

The analytical notes answer "what *is* the least-complexity network, as a variational object?" This report answers the operational questions that follow: which of those constructions can be computed; what a faithful numerical surrogate looks like when the exact object is intractable; and what the resulting meta-optimizer actually selects when pointed at data. The guiding discipline throughout development was conservative: one capability added at a time, each premise checked before the capability was built on top of it, negative results recorded as first-class outcomes rather than rationalized away, and nothing consolidated into the package until it passed a regression check. This report reflects that discipline—in particular, it distinguishes carefully between the components that are *exact* realizations of the theory, those that are *approximate* surrogates, and those that remain *unrealized*.

All quantitative results in §11.1–§11.3 were regenerated from the current package state using the reproducible runner scripts shipped in `validation_runners/` (§12); no number is carried over from memory or an earlier package version. Small run-to-run variation (seed, epoch budget, machine) affects the third decimal of accuracies but not the selected architectures, which are stable.

## 2 From analytical formulation to numerical implementation

The analytical framework is a nested hierarchy: an inner loop that optimizes weights at fixed architecture, and outer loops that optimize width, depth, and—ultimately—the primitive operation itself. Table 1 records how each analytical formulation was mapped to code, and the subsections that follow expand on the entries that required approximation or were left unrealized.

### 2.1 What is realized exactly

Three components are faithful, not approximate. *Weight optimization* is ordinary backpropagation; the analytical reading of it as the forward/adjoint legs of a discrete least-action problem is an interpretation, and the computation is exactly standard training. *Width sparsity* is realized as the theory specifies: the total-variation dual certificate and a column-generation / conditional-gradient solver terminate at a finite atom (neuron) count controlled by the regularization price $\lambda$, matching the finite-atomicity proposition of the two-layer theory. *Criticality* is realized as an exact mean-field diagnostic: the slope $\chi_1 = \sigma_w^2\,\mathbb{E}[\sigma'(z)^2]$ is computed in closed form, the critical coupling is located (for tanh with $\sigma_b^2 = 0.05$, at $\sigma_w^2 \approx 1.76$), and the recurrent primitives are initialized there. These are the components inherited most directly from the two-layer core, where the theory is exact.

### 2.2 What is realized approximately

The depth formulations are the clearest case of principled approximation. Priced depth (Route 2) is implemented as a marginal-value stopping rule over integer depth, but the theory's headline corollary—$L^\star \sim \mu^{-1/(\alpha+1)}$, the closed-form coupling of optimal depth to the depth price through the critical exponent—rests on a heuristic marginal-mode-dominance argument. It was validated only on synthetic ground truth; on the compact *sequence* datasets examined, optimal depth was $L^\star = 1$ (depth-limited), consistent with but not a strong test of the scaling (deeper optima do appear on relational tasks—ESOL selects $L^\star = 4$, §7.2). Geometric depth (Route 1) is implemented as a continuous-depth neural-ODE [5] flow whose complexity is arc length in a learned metric; the prediction that a ReLU flow can be arbitrarily deep at finite complexity cost was tested by measuring arc length as a function of integration time at matched criticality and was *not confirmed*—the flow settled to approximately constant speed, giving arc length linear in integration time for both smooth and ReLU vector fields. The mean-field / Wasserstein dynamics are realized only implicitly: the network is trained by gradient descent on its finite parametrization, and the measure-space flow is a lens on that process rather than a separately integrated object.

### 2.3 The architecture parameter, and the one genuine cross-contract seam

The architecture parameter $\boldsymbol{\alpha}$ selects *between* computational graphs. Within a contract it is not a bolt-on: §9.2 derives it as the outer Gibbs measure of an entropic free energy on the primitive simplex, with DARTS-style softmax training its Fisher–Rao gradient flow and the co-adaptation-free solo marginal risk the forced energy. What remains genuinely irreducible to the continuum machinery is only the *discrete cross-contract dispatch* of §4.2: distinct contracts have incompatible interfaces, so the choice between them cannot be relaxed to a simplex and is settled by a minimum-description-length (MDL) bake-off rather than a gradient flow. (Earlier iterations framed $\boldsymbol{\alpha}$ itself as an unreconciled "bolt-on"; that framing is superseded by the derivation, with a short historical note recorded at the end of §9.2.)

### 2.4 What is not realized

True architecture-invariance—an optimizer that can invent a primitive outside the vocabulary —is not implemented, and by the analytical argument cannot be, without surrendering the continuum machinery: it is the uncomputable outer problem the tractable framework was built to avoid. The explicit Wasserstein measure flow, as noted, is not integrated. And the group parameter of the weight-sharing primitive (which symmetry to share under) does not relax smoothly; the implementation sidesteps this by instantiating the two useful groups (translation, time) as separate selectable primitives rather than searching over groups.

## 3 The processing pipeline: a physicist's ordering

This report is organized around the structure the code actually has: a single processing pipeline whose stages follow the causal sequence a physicist imposes when writing down an effective theory. Fix the *kinematics* (the contract and its symmetries) first, then count the *degrees of freedom* (how many modes the contract carries), then solve the *dynamics* (train the couplings), and only then read off the *observables*. Every switch the package exposes falls into exactly one of these stages, and the sections that follow are grouped by stage: *kinematics* covers contract dispatch (§4.1–§4.3), tensor-structure discovery (§5), and symmetry discovery and canonicalization (§6); *degrees of freedom* covers the priced width/depth/mode selection (§7); *dynamics* covers the primitive vocabulary and the derived, priced architecture parameter $\boldsymbol{\alpha}$ (§8.1–§9.4); and *observables* covers the primitive readout and interpretability (§10). Validation is consolidated afterward (§11.1–§11.5).

In the implementation this ordering is realized as a single `fit(select_size=True)` flow rather than a set of detached pre-steps: routing and canonicalization (kinematics) complete first; the width/depth selection (§7) runs *in-line* next, sizing the network for the contract that was just fixed; training follows; and the readout (§9.2) discretizes the trained mixture last, guarded by `hasattr(.cells)` so it is a safe no-op on the generated-equivariant contract. The MDL action $J = R + \mu\,\Omega$ is not a stage in this sequence but the object minimized *at* each selection.

**The ordering is load-bearing: two bugs it exposed.** Enforcing the sequence is not cosmetic. Running the full stack surfaced two defects that are, at root, *ordering violations*, and fixing the order fixed them. First, canonicalization augments the node features with principal-axis coordinates, changing their dimension; when the degrees-of-freedom sweep ran *before* that augmentation it sized the network for the wrong input width, producing a shape mismatch at train time. Canonicalization is a quotient of the configuration space—kinematics—so it must precede mode-counting; moving it above the sweep resolves the crash. The same quotient must also be applied to held-out data at inference, or the deployed network (trained on augmented features) receives the wrong dimensionality; a stored `apply_canonicalization` transform, reused at test time, restores train/inference consistency. Second, the rotation-invariance test that decides whether to canonicalize compared the invariant-feature fit against the orientation-feature fit with no absolute floor. On a target that is unpredictable at the current budget—both fits negative—it would still declare "rotation-invariant" (the larger of two failures) and canonicalize onto the set contract, discarding the equivariant contract that genuinely fits. An absolute floor `min_invariant_r2` plus an *uninformative-defer* path (§6) corrects this: when neither fit clears the floor the test abstains and routing is left to the structural default, which keeps the geometry. On rMD17-ethanol this lifted the full-stack result from

$R^2$ 0.014 to 0.51. A parallel guard was needed one rung down: the width/depth sweep itself, when run at a smoke-budget number of epochs, sees a flat noise curve in which no width has trained enough to reveal its capacity, and selects the mode count essentially at random. Flooring the *sweep* epochs (so the measurement reaches the signal regime, independent of the deployment budget) and adding an uninformative-sweep guard (fall back to maximum capacity when even the best width fails to clear a fit floor, rather than prune on noise) turns the rMD17 width curve monotone and selects a principled $K^\star = 64$; the end-to-end result rises to $R^2$ 0.586. The recurring lesson is that a capacity or symmetry decision is only meaningful once the measurement it rests on has left the noise regime—the analogue of not reading mode occupancies off a configuration that has not yet thermalized.

**Per-layer robustness and cost.** To turn these switches into a usable default, each was ablated individually against a plain baseline across nine compact real datasets spanning four data types—1D sequence (ItalyPowerDemand, ECG5000, ArrowHead), 2D image (a BloodMNIST subset, an MNIST subset), geometric sets (two JetNet pairings), a molecular graph (ESOL), and a rotation-invariant equivariant target (QM7)—with up to five seeds per (dataset, layer) to estimate variance. The layers separate cleanly into three groups. The *readout* layers are the only cheap ones that actively move results: the sparsity-priced readout never hurt on any of the nine datasets and reduced seed variance on the noisy ones at unit cost, while the Gibbs readout is a conditional tool—it roughly halved seed variance on noisy geometric tasks (an 82% variance cut on QM7 at essentially no cost to the mean) but was a net negative on clean data, confirmed on all five clean datasets. The *routing and contract* layers (tie-break, symmetry routing, canonicalization, discovery) are correct no-ops whenever the default router already respects the data's structure, which is the common case; their value appears specifically on inputs the default would mis-route (the rMD17 canonicalization rescue), and the tie-break can even slightly backfire on clearly-geometric data where the default is already right. The degrees-of-freedom sweep is the highest-value single layer where it applies—best skill *and* best robustness on the graph-regression task—but its cost, dominated by the epoch floor that makes its measurement trustworthy, ranges from a 10× overhead to cost-prohibitive on large sets.

**Three presets.** These findings are packaged as an evidence-based `--preset` switch on both validation runners, with explicit flags overriding the preset. MIN is the plain baseline (argmax readout, all structural layers off): cheapest, and best on clean data where every added layer was inert or slightly harmful. MED is the "only-ever-helped" set—the sparsity-priced readout plus the routing/contract layers as near-free insurance (correct no-ops when unneeded, high-value on mis-routed geometry)—and deliberately excludes the Gibbs readout (harmful on clean data), the tie-break (can backfire on geometric data), and the degrees-of-freedom sweep (expensive). MAX engages every layer (Gibbs readout, tie-break, symmetry routing, canonicalization, discovery, and the width/depth sweep): the most robust configuration on noisy data with genuine signal, at the highest cost, with the sweep occasionally prohibitive on large set data. The presets make the recommended default (MED) a single flag while keeping the full switch space available.

# 4 Contracts and dispatch

The outermost pipeline rung fixes the *contract*: which contract the datum lives in and under which symmetry group the operations must be equivariant. This section defines the eight contracts and their vocabularies (§4.1), the meta-router that discovers which contract a datum satisfies and dispatches to it (§4.2), and the description-length objective that folds the near-tie contract choice into the same $\mathcal{J} = R + \mu\,\Omega$ minimization as every inner rung (§4.3). The two remaining kinematic operations—discovering latent tensor structure and discovering symmetry—follow in §5 and §6.

### 4.1 The eight computational contracts

The unified sequence schema (§8.3.1) was the first of eight peer schemas, one per computational contract. A contract is a fixed way of presenting the datum together with the symmetry group under which the operations must be equivariant; the six irreducible primitives of §8.1 are realized afresh in each contract under its group action and tensor rank. Table 2 inventories them.

**Why the contracts must be separate.** The grid ladder cannot be collapsed: an $N$-dimensional convolution is not expressible as a stack of lower-rank ones, so each rank needs its own contract. The equivariant contract cannot be folded into the plain graph contract because its features are steerable (a direct sum of a rotation-invariant scalar part and a rotation-covariant vector part), and one cannot add a type-1 output to a type-0 output, nor apply an ordinary ReLU to a vector feature without breaking equivariance; equivariant primitives require per-irrep mixing and norm-gated nonlinearities. The set contract is the maximal-symmetry endpoint: graph aggregators collapse or crash on an edgeless collection, and attention-based set blocks (SAB) outperform sum decomposition exactly when higher-order interactions matter. Consistency across the eight was enforced mechanically: every top-level class exposes `forward/architecture/alpha_report`, every cell exposes a `mixed` method, and every builder takes the canonical $(n_{\text{in}}, \text{width}, \text{depth}, n_{\text{out}}, \text{primitives})$ signature, with a self-check that fails if any of these regress.

### 4.2 The meta-router: one entry point over eight contracts

The eight schemas are unified behind a meta-router, `AllGraph`, whose single `fit(data)` entry point discovers which contract the data satisfies and hands off to that schema's own primitive/width/depth search. Because the contracts have incompatible interfaces there is no shared tensor space in which to mix them, so the router is a two-level *dispatch*, not a differentiable mixture (the structural reason is developed in the companion analytical report).

**Level 1: structural type dispatch.** The data container decides the contract with certainty: an edge set routes to the graph contract (or, with positions also present, the equivariant contract); a variable-size collection with no edges routes to the set contract; a fixed-shape array routes to the grid family. This level is rule-based and overridable—the user can force a contract—because the container is itself the type signal.

**Level 2: grid-rank dispatch.** Within the grid family the tensor rank fixes the member (rank-3 $\to$sequence, 4 $\to$spatial, 5 $\to$volumetric, 6 $\to$ 4D); a flat feature vector is sent to the mutual-information mode-structure test of §5, which recovers a latent 1D or 2D lattice. Level 2 thus reuses the representation front-end wholesale.

**The learned tie-break.** The one genuinely ambiguous case is geometric data (coordinates present), which may be modelled as an equivariant graph, a plain graph, or a bare point set. Since the contracts cannot be mixed, the router runs a *bake-off*: each admissible contract is trained briefly from scratch on a held-out split and ranked by the MDL objective $\widehat{R}_c + \mu_c\Omega_{\text{struct}}(c)$—the clean-solo protocol of §9.1 lifted from primitives to whole contracts, and folded into the same $R+\mu\,\Omega$

objective as every inner rung (§4.3). The charge $\Omega_{\text{struct}}$ is a *derived* structural code length (not an ordinal fiat), so a richer contract must earn its cost. On controlled targets this selects correctly in every case: a geometry-dependent target (radius of gyration) selects equivariant decisively; a purely topological target (mean degree) is a near-tie in fit between graph and equivariant, which the structural charge breaks toward the cheaper graph; a pure set statistic selects the set contract outright—itself a near-tie with equivariant that the charge resolves. The tie-break is the empirical, across-contract analogue of the least description-length rules that govern the within-contract search.

**End-to-end behaviour.** Routed through `AllGraph.fit` on their data containers alone, all real datasets dispatched to the correct schema, and two of them produced genuinely *heterogeneous* per-layer architectures on real data—the ESOL graph task discretizing to `gin`→`pna`→`pna` and the rMD17 equivariant task to `e_painn`→`e_painn`→`e_tp`, the latter mixing the two equivariant message-passing philosophies across depth. This settles, affirmatively, the depth-1 question left open in the first report (§7): given a real task with compositional structure and a deep-enough contract, the metaoptimizer does select distinct primitives at distinct layers.

### 4.2.1 Amortizing the bake-off: learned, transferable contract routing

The bake-off is correct but expensive—it trains two or three whole contracts from scratch for every new geometric dataset—and it is *not transferable*: nothing learned on one dataset speeds up the next. Yet a practitioner's prior ("a molecular force field wants equivariant; a topological property wants a plain graph; a bag of atoms wants a set") is exactly the kind of regularity a cheap meta-model should capture. We therefore added a learned router (`machinery/learned_contract_router.py`) that predicts the contract the bake-off *would* choose from training-free dataset descriptors—a surrogate for $\arg\min_c[\widehat{R}_c + \mu_c\Omega_{\text{struct}}(c)]$ that predicts the argmin without computing each $\widehat{R}_c$. It is the same MDL objective, amortized.

**The descriptor.** The discriminative signal is *what the target depends on*, captured by three proxy-correlations of $y$ with structure-specific, permutation- and rotation-invariant descriptors: a geometry proxy (correlation of $y$ with a pairwise-distance summary), a topology proxy (with an adjacency-spectrum-plus-degree summary), and a set proxy (with node-feature moments). On controlled targets the winning proxy always matches the true contract (geometric 0.98/0.20/0.08, topological 0.23/1.00/0.08, set 0.05/0.09/1.00), with no schema training.

**The model and its correctness floor.** A nearest-prototype classifier in standardized descriptor space, with confidence read off the nearest-versus-runner-up margin. The policy has a correctness floor: if the router is confident, its prediction is used and the bake-off is skipped; if not, the bake-off runs (the ground truth) and its outcome is recorded to improve the router. So the router is never worse than the bake-off and much cheaper when confident. Prototype leave-one-out transfers 12/12 across dataset scales on controlled targets.

**Full-budget labels: the router is only as good as its labels.** The router learns from bake-off outcomes, so a mislabelled bake-off teaches a wrong mapping. A premise check found the bake-off winner *flips with budget* on the two real geometric datasets: at the tiny quick-validation tie-break budget the equivariant branch has not converged, so the $\Omega_{\text{struct}}$ charge spuriously prefers the cheap set/graph contract, whereas at an adequate budget equivariant wins decisively and its advantage widens (rMD17 $R^2$ climbs $0.06 \to 0.20 \to 0.44$ across 15/40/90 epochs while set and graph stay near zero; QM7-equiv 0.62 → 0.72). A dedicated recorder (`validation_runners/record_contract_corpus.py`) runs the bake-off at an adequate budget on every geometric dataset and persists the (descriptor, winner) corpus. Two guards in `tiebreak` keep reduced-budget runs from corrupting this: outcomes are recorded only when the budget clears a trust threshold, and when the fallback bake-off would itself be low-budget the router's full-budget-seeded prediction is preferred over it. This is the honest resolution of the reduced-versus-full-budget tension: full-budget knowledge lives in the router; a cheap bake-off neither overrides nor poisons it.

**Transfer across physical domains, and a descriptor lesson.** Broadening the corpus beyond chemistry to particle-physics point clouds (JetNet jets: jet width → equivariant, total transverse momentum → set) exposed a real transfer failure and its fix. With the full descriptor, cross-domain leave-one-out was only 4/7, because the raw size statistics (node and edge counts) live on incomparable scales across domains—a molecule's handful of bonds versus a cloud's many proximity edges—and in the standardized distance they swamp the discriminative proxies. Matching on the *dimensionless* proxy-correlations alone (each a correlation in $[0, 1]$, hence domain-portable) lifts cross-domain leave-one-out to 6/7 with no within-domain regression (15/15); the full descriptor is still stored but matching uses the transferable subset. The one residual miss (a jet-width target with both geometry and topology proxies high) is a genuine geometry–topology coupling, the same honest ambiguity as QM7 energy.

**A negative result on descriptor enrichment.** Because QM7-equiv's proxies are genuinely mixed (geometry 0.62, topology 0.72), we tested whether richer features sharpen the distinction. They do not. Spectral features (distance-matrix spectrum for geometry, graph-Laplacian spectrum for topology) *widen* no margin: on QM7 both spectra reach ∼0.95 because atomization energy really does depend on both geometry and topology (bonding fixes geometry), and on a ring-versus-chain target the crude degree statistic already beats the Laplacian gap. Persistence-homology features ($H_0/H_1$ of the Vietoris–Rips filtration) likewise lose to crude distance moments on a ring-versus-blob target, since a ring and a blob already differ in distance variance. The conclusion, recorded as a first-class negative result, is that crude proxies are already sufficient wherever a separable signal exists, and the genuinely mixed cases are correctly handled by confidence-deferral, not by richer descriptors—enrichment adds cost (eigendecompositions, a persistence dependency) with no measured benefit, so the descriptor is left as the validated crude vector.

**End-to-end effect.** With the learned router active (default, warm-seeded from the cross-domain corpus) the two geometric molecular datasets route to the *equivariant* contract in ∼0.2 s with the bake-off skipped, and the quick-validation skill improves markedly relative to the mis-routed baseline: QM7-equiv $R^2$ 0.57 → 0.77 and rMD17 $R^2$ −0.00 → 0.15, the latter moving from below the trivial baseline (a geometry-blind set contract) to genuinely predictive SO(3)-equivariant message passing (`e_painn`→`e_painn`→`e_tp`). Disabling the router (`--no_learned_router`) reproduces the old low-budget mis-routing to set, which is the controlled demonstration that the learned, full-budget-seeded router is what supplies the correct contract cheaply.

## 4.3 Folding the contract choice into the MDL objective

The choice between contracts near a fit tie is folded into the same $\mathcal{J} = R + \mu\,\Omega$ objective as width, depth, and primitive, using a *derived* structural code length $\Omega_{\text{struct}}(c)$ (`machinery/contract_mdl.py`). This replaces the single hand-set ordinal charge $\kappa(\text{set}) < \kappa(\text{graph}) < \kappa(\text{equivariant})$ that earlier iterations used to break near-ties in the meta-router (§4.2)—the one hand-set complexity number that had remained in the framework, now derived rather than posited.

**Why discrete is not an obstruction.** The interface incompatibility that makes the contract a *dispatch* rather than a mixture (§4.2) does not also foreclose *pricing*: like depth—already a discrete rung selected by a marginal-value rule—a discrete choice can still be minimized under $R+\mu\,\Omega$ given an ordering and a per-step code length. So the fold needs only a principled $\Omega_{\text{struct}}(c)$.

**The structural code length.** $\Omega_{\text{struct}}(c)$ is the description length of the scaffolding a contract commits to, per datum, as excess over the cheapest contract. For a datum with $N$ units and $E$ edges in $d$ dimensions,

$$\Omega_{\text{struct}}(\text{set}) = 0, \qquad \Omega_{\text{struct}}(\text{graph}) = \log\binom{N(N-1)/2}{E} \approx E\log\tfrac{N^2}{2E},$$
$$\Omega_{\text{struct}}(\text{equiv}) = \Omega_{\text{struct}}(\text{graph}) + \left(Nd - \tfrac{d(d-1)}{2}\right)\log\tfrac{1}{\delta},$$

the bits to write down the adjacency (topology) and then the geometry ($Nd$ coordinates at resolution $1/\delta$ minus the $\dim \mathrm{SO}(d)$ gauge the equivariance quotients out); a grid-rank tower (sequence < spatial < volumetric < 4D) is charged analogously by rank plus an axis refinement. This is strictly monotone up the structure-inclusion lattice and data-dependent. Crucially it prices *structural* richness, not parametric complexity: a $\frac{1}{2}k\log n$ (parameter-count) charge *misorders* contracts, since an equivariant model can be parameter-cheap—steerable features are compact—yet structurally over-rich for a target that does not use geometry. Parameter count stays the within-contract currency (width, depth); the contract's own charge is structural.

**Selection and validation.** The primary selector `select_contract_mdl` minimises the global $\mathcal{J} = \widehat{R}_c + \mu_c\Omega_{\text{struct}}(c)$ over all admissible contracts; a marginal-value form (`marginal_value_contract`) is retained as a certificate reading ("does climbing the lattice pay?"), but the *global* objective is primary because the greedy marginal rule under-climbs on a non-monotone fit profile—on a geometric target the set→graph step adds no fit (edges do not help a coordinate quantity), stopping the greedy climb before the payoff at equivariant two steps up. The global $\mathcal{J}$, like the depth frontier, is robust to this. Wired into `AllGraph.tiebreak` with $\Omega_{\text{struct}}$ computed from the actual node and edge counts, it reproduces the physically correct contract on all three controlled targets (Table 3).

With this the entire complexity ladder—contract, primitive, depth, width, weights—is one MDL functional $R+\mu\,\Omega$, discrete at the top two rungs (contract, and the primitive $\arg\max$) and continuous below, with every $\Omega$ a derived code length rather than a hand-set constant.

### 4.4 The functional code length: pricing the effective degrees of freedom of the fit

$\Omega_{\text{struct}}(c)$ prices the *scaffolding* a contract commits to. It says nothing about how many degrees of freedom the *fitted function* actually uses once trained inside that scaffolding. Singular learning theory supplies the missing term. A recent result—minimum description length meets singular learning theory [26]—is that the stochastic code length of a trained model is $\mathrm{MDL}_{\text{sing}}(D) = -\log p(D \mid w^\star) + \lambda\log n + \text{const}$, with $\lambda$ the local learning coefficient (the RLCT) [15, 27]. This is *exactly* the package's $\mathcal{J} = R + \mu\,\Omega$ with the residual playing $R$ and a *functional* code length $\omega_{\text{func}} = \lambda\log n$ playing the complexity charge—the same singular free energy $F_n = nL(w^\star) + \lambda\log n$ that anchors the thermodynamic potential of §9.3, now read as a per-contract description length. Where the parameter count $\frac{1}{2}k \log n$ *misorders* contracts (a steerable equivariant model is parameter-cheap yet may be functionally simple or complex depending on what it fits), $\lambda \log n$ with $\lambda \leq k/2$ prices the effective, not the nominal, dimensionality.

The package makes this an opt-in refinement of the contract charge (`machinery/singular_mdl.py`, `price_singular`): for each admissible contract whose candidate net is trained during the bake-off, the local learning coefficient is estimated at its optimum by SGLD [15] (`singular_complexity_of`), and the contract's total charge becomes $\Omega_{\text{total}}(c) = \Omega_{\text{struct}}(c) + (\mu_s/\mu_{\text{c}})\,\omega_{\text{func}}(c)$ before the same normalization and arg min. Two guards keep a noisy estimate from corrupting the selection: an RLCT is non-negative, so a slightly negative $\hat{\lambda}$ (SGLD noise near a shallow basin) is clamped to zero—negligible functional complexity, never a negative code length—while a *strongly* negative $\hat{\lambda}$ (a non-converged optimum, where the estimator is meaningless) fails a validity check and the contract keeps structural-only pricing. A garbage $\lambda$ therefore never enters $\mathcal{J}$.

The honest scope of this term is precise, and two studies bound it. On QM7-equivariant data (real geometry) the functional charge *fires and guards* correctly—the converged set contract is priced $\omega_{\text{func}} \approx 4$ nats, an under-converged equivariant candidate is declined by the guard—but it does not change the winner, because the structural spread (set 0 / graph $\sim$32 / equiv $\sim$127 nats) dwarfs it. The sharper question—whether $\omega_{\text{func}}$ can ever be *decisive*—has a clean answer. A flip requires two contracts tied in *both* structure and fit yet converging to functions of different singularity. Constructing that tie on complete-graph data (where $\Omega_{\text{struct}}(\text{graph}) = E \log \frac{P}{E} \to 0$ as $E \to P$, matching set at 0 nats) and measuring $\lambda$ at convergence across seeds and prices, natural training *never* flips: at a structure-and-fit tie the simpler class (set) simultaneously wins the structural tie-break *and* carries the lower $\lambda$ (it fits as well with fewer effective degrees of freedom), so the two Occam pressures align and $\omega_{\text{func}}$ can only reinforce, never overturn, the structural verdict. The term *is* correctly signed and *would* bite in the anti-correlated corner—if the fit-leader were the more singular contract, a $\sim$1.5-nat functional gap flips the choice at $\mu_{\text{c}}$=0.05—but that corner is not reached on realistic converged data. The functional code length is therefore a faithful, guarded diagnostic of effective degrees of freedom and a principled refinement of $\mathcal{J}$; it is second-order to the structural charge in every natural regime, which is precisely why fusing it is safe.

# 5 The representation front-end: discovering tensor structure

The metaoptimizer of §7 selects a primitive, width, and depth *within* a fixed input representation—sequence data enters as $(b, T, c)$, images are flattened, and so on. But the representation is itself a choice, and often the most consequential one: a $28 \times 28$ image forced through a recurrent sequence model is being asked to discover its own two-dimensional adjacency from scratch. Before selecting operations, then, a physicist asks a prior question—what *shape* does the data naturally have?—and answers it from the data's correlation structure. This section documents that stage. It sits upstream of both the size metaoptimizer (§7) and the symmetry front-end (§6), and completes the pipeline: discover the tensor structure, tensorize accordingly, (optionally) quotient symmetries, then search for a minimal model.

## 5.1 Mode-coupling structure from mutual information (MI)

The organizing idea, borrowed directly from tensor-network structure search, is that the right representation is the one whose correlation (entanglement) structure matches the data's: a one-dimensional chain (matrix-product-state-like) suits data with band-diagonal, distance-decaying correlations, while a two-dimensional grid (projected-entangled-pair-state-like) suits data whose correlations decay with a *2D* grid distance. We make this operational (`core/mode_structure.py`) by computing the pairwise mutual-information matrix $M_{ij}$ between input coordinates (a Gaussian proxy $M_{ij} = -\frac{1}{2}\log(1-\rho_{ij}^2)$ for speed, or a binned estimator), then scoring how well $M$ is explained by 1D-chain adjacency versus each candidate 2D grid tensorization—correlating $M_{ij}$ against negative coordinate distance under each hypothesis. The best-fitting hypothesis wins only if it clears an absolute floor and a margin over the runner-up; otherwise the verdict is *unstructured*. As with the symmetry detectors, the absence verdict is a first-class output: the method must not hallucinate structure that is not there.

On known controls this recovers the truth cleanly: a 1D autoregressive series is classified 1D (fit 0.72); a synthetic $6 \times 6$ grid field is classified 2D and—notably—its *exact* $6 \times 6$ shape is recovered (fit 0.71, beating both the 1D hypothesis and the wrong reshapings $4 \times 9$, $9 \times 4$, $3 \times 12$); and both a shuffled grid and pure noise are correctly called unstructured (fit 0.11 and 0.05, below the floor). The verdict also carries an actionable recommendation—the class of primitives to search: sequence primitives for 1D, the conv2d family for 2D, dense/attention for unstructured.

### 5.2 Exact tensorization via the stride-anomaly signal

Classifying *1D-versus-2D* is not enough to feed a convolution; we need the exact $H \times W$. The coarse MI-versus-distance fit is too blunt for this on natural images—it collapses the whole grid geometry into one scalar and, on CIFAR-10, preferred a spurious $64 \times 16$ over the true $32 \times 32$. The sharper signal is structural: in a row-major $H \times W$ flattening, vertically-adjacent pixels sit *exactly* $W$ *apart* in the flat index, so the MI profile as a function of flat-distance shows a peak at $\delta = W$ that *breaks* the otherwise-monotone decay. We score each candidate width by how much the MI at that stride exceeds the smooth decay trend interpolated from its neighbours, $p[W] - \frac{1}{2}(p[W-2] + p[W+2])$. A genuine 2D grid produces a clear positive anomaly at the true $W$; a 1D chain has monotone decay and hence no positive anomaly, so it is correctly never promoted to 2D.

This estimator is exact where the coarse fit failed. On CIFAR-10 the MI profile climbs $0.46 \to 0.68 \to 0.93$ across $\delta = 30, 31, 32$ and falls back—an unambiguous stride at 32—and the recovered shape is $32 \times 32$. On Fashion-MNIST, a *second, independent* image dataset, it recovers $28 \times 28$ exactly. Synthetic grids ($6 \times 6$, $4 \times 9$, $8 \times 4$) are recovered exactly, the 1D control stays 1D, and time series (GunPoint) stay 1D. The method also extends to three dimensions: a synthetic $4 \times 4 \times 4$ volume shows the two-peak signature (strides at $W = 4$ and $W \cdot H = 16$), and real CIFAR colour, a genuine three-tensor, shows peaks at both 32 (within-row) and $1024 = 32 \times 32$ (the colour-channel jump). This 3D signature, which originally rested on a synthetic volume, is now validated on real volumetric data and priced across ranks 1–4 in §5.3.

### 5.3 Priced N-D tensorization: rank 1–4 and the real-data assessment

Two limitations of the stride-anomaly estimator above were subsequently closed. First, its acceptance was a hand-set floor-plus-margin rule; second, it reported only the leading 2D stride, and its 3D signature rested on a single synthetic volume. Both are resolved by folding tensorization into the same $\mathcal{J} = R + \mu\,\Omega$ objective as every other rung (`machinery.contract_mdl.price_tensorization`, `mode_structure.parse_grid_shape`), extended to any rank $1 \leq r \leq 4$—the latter covering the 4D schema directly.

**Position-aware fit, priced rank.** The flat-distance profile averages $M_{i,i+\delta}$ over all positions, which washes out at axis boundaries and fails on length-2 axes. The robust successor scores a candidate shape by the *position-aware* mean mutual information over its *true* axis-neighbour pairs (multi-index differing by one in exactly one axis). Selection is two-stage: within a rank the best factorisation maximises this neighbour MI; across ranks, the smallest rank whose neighbour-MI risk is within tolerance of the minimum is chosen—Occam on rank, the marginal-value rule one level out. A *significance gate* guards against finite-sample MI bias: since every MI estimate is positive-biased by $\sim 1/n_{\text{samples}}$, a shape is promoted beyond 1D only when its neighbour MI exceeds the mean off-diagonal MI by a factor (default 1.6). A real lattice has neighbour pairs far more correlated than random pairs (ratio $\sim 5$ on images); pure noise has them equal (ratio $\sim 1$).

**Real-data assessment.** The estimator was evaluated on real datasets with known ground-truth shape (Table 4), the basis for enabling it as an allgraph default. It recovers real 2D images (MNIST, BloodMNIST) with decisive margins, keeps real time series and unstructured noise at 1D, and recovers real 3D volumes (OrganMNIST3D) when the volume reads as a clean grid. Crucially, every failure mode is *fail-safe*: the estimator under-ranks to a coarser structure (e.g. a whole-anatomy strided downsample that ties at rank 1) rather than imposing a false lattice.

Because the estimator only ever *refines* structure it is confident in and otherwise leaves the input as routed, it is enabled by default in the meta-router (`tensorize_mu`$= 0.05$): turning it on cannot degrade routing, only upgrade a genuinely grid-shaped flat vector to its matching schema.

**Scalability: on-demand MI for full-resolution volumes.** The dense estimator forms the full $D \times D$ mutual-information matrix, which is $\sim 4$ GB at $D = 21952$ (a $28^3$ volume) and intractable. This is resolved by the observation that the position-aware scorer only ever reads $M$ at $O(D)$ axis-neighbour pairs per candidate and the significance gate needs only a *sampled* off-diagonal mean—so the full matrix is never required. The scalable parser (`mode_structure.parse_grid_shape_scalable`) computes each pair's MI on demand over a sample, with the candidate set bounded to the divisor lattice (the role of $\Omega_{\text{struct}}$ here: it prices and caps rank, so factorisations are enumerated rather than searched unboundedly). It is *identical in logic* to the dense parser and *agrees with it on every size where both run* (7/7 shapes, exact match), runs a full $28^3$ volume in $\sim 40$ s (versus out-of-memory, OOM), and scales roughly linearly in $D$ rather than quadratically. It is auto-selected above $D = 500$. The residual honest gaps are now only the two documented signal-strength limits—background-dominated or exotic-aspect-ratio volumes may under-rank—and both are fail-safe, never mis-ranked.

### 5.4 Routing: automatic tensorization end to end

The detector's verdict drives a router (`core/route.py`) that dispatches each dataset to the schema whose tensor interface matches and reshapes the flat input accordingly: a 2D verdict goes to the spatial (conv2d) schema on the discovered $H \times W$; a 1D verdict to the sequence schema; an unstructured verdict to the sequence schema as a length-1 dense vector. This turns the representation decision into an automatic, data-driven stage. Two implementation lessons were recorded during validation. First, the spatial schema's flatten-`dense` primitive ($\sim 10^6$ parameters) overfits catastrophically and defeats the purpose of routing to a spatial model at all; it is excluded from the spatial recommendation, leaving the genuinely spatial primitives (conv2d, depthwise conv, pointwise, attention, norm). Second, a per-pixel standardization with too small a variance floor amplified near-constant border pixels into a train/test scale mismatch that collapsed test accuracy to chance—a reminder that preprocessing bugs masquerade as modelling failures. Both were fixed before the comparison below.

### 5.5 Head-to-head: does automatic tensorization help?

We compare, on matched autonomous width/depth/primitive selection, a *fixed* pipeline (every dataset a sequence) against the *routed* pipeline (auto-tensorized to the matching schema). Crucially, the width and depth here are chosen by the theoretical marginal-value rule of §7, not a grid search: layers are added only while their per-layer marginal loss reduction exceeds the price $\mu$, so the selected depth tracks the marginal-versus-price comparison per dataset. On Fashion-MNIST the marginal from depth one to two (0.20–0.26) far exceeds $\mu = 0.02$, so a depth-2 spatial model is selected; on GunPoint the marginal is near zero, so depth one is kept; on the tabular series the second layer is added but the third (negative marginal) is not. The routing results, with the selected architectures, are:

| Dataset | detected | fixed (seq) | routed |
|---|---|---|---|
| Fashion-MNIST | 2D $28 \times 28$ | `[plain,linssm]` 0.405 | `[conv_dw`$\times$`2]` spatial **0.640** |
| ItalyPowerDemand | unstructured | `[spectral]` 0.632 | `[norm`$\times$`2]` dense-vec **0.948** |
| GunPoint | 1D 150 | `[gated]` 0.567 | `[plain]` sequence 0.560 |

The pattern is the intended one. Where the data has genuine non-sequence structure, routing helps substantially: Fashion-MNIST gains +0.235 by going to the spatial (conv2d) schema, and the tabular series gains +0.316 by being treated as a dense vector rather than a spurious sequence. Where the data is a genuine sequence, routing correctly keeps it a sequence and does no harm (GunPoint, −0.007, within noise). The architecture the metaoptimizer selects changes class accordingly—a recurrent cell on the forced sequence becomes a depthwise convolution on the routed image—so routing improves both accuracy and architectural appropriateness. We note the compute caveats plainly: spatial grids were pooled and sequences capped for tractability, so absolute image accuracies would rise at full resolution; the routing *comparison*, not the absolute number, is the object of interest. Symmetry preprocessing (§6) is available as an opt-in stage in this pipeline but, per the project's policy, is off by default.

### 5.6 A note on the selection mechanism

Because §7 and this section both speak of "selecting" width and depth, it is worth stating precisely what that means, as the framework contains three distinct mechanisms and they are not interchangeable. The first is the *priced marginal-value rule* (`machinery/priced_depth.py`): it measures the depth curve $S^\star(L)$, forms the per-layer marginal $-\partial S^\star/\partial L$, and stops at the last layer whose marginal exceeds the price $\mu$. This is not an accuracy argmax—on a diminishing-returns curve where a grid search would pick the deepest model, the priced rule selects $L^\star = 1, 2, 3, 4, 5$ as $\mu$ falls through $0.5, 0.2, 0.1, 0.02, 0.005$, tracing the fit–complexity frontier. The second is the *differentiable complexity penalty* (§7, the benchmark of record), which folds $\mu \cdot$cost into the architecture objective so gradient descent on $\alpha$ is pulled toward cheap primitives during search. The third is plain grid search with accuracy compaction, used only where the question under study is the representation rather than the selection rule. The first two are the theory-grounded realizations of the action $J = R + \mu\,\Omega$; only they should be read as tests of the metaoptimization principle.

## 6 The symmetry-discovery pipeline: a physicist's front-end

The work of §1–§12 optimizes architecture *within* a fixed input representation. A separate line of development asks a prior question, in the spirit of how a physicist approaches a new system: *before* fitting a model, can we discover the symmetries of the data, quotient them out to reduce the problem's dimensionality, and only then search for a minimal model on the reduced representation? This section documents that pipeline. It is a representation/preprocessing stage that runs *before* the metaoptimizer of the preceding sections; conceptually it is the outermost, cheapest rung of the same minimum-description-length ladder the analytical notes develop (width sparsity removes redundant neurons; symmetry-quotient removes redundant coordinate directions; the information bottleneck removes all task-irrelevant information—§6.4).

### 6.1 Continuous symmetry discovery via the Lie-derivative nullspace

A continuous (Lie) symmetry with infinitesimal generator $L$ leaves a function $f$ invariant iff $\nabla f(x)^\top(Lx) = 0$ for all $x$. Stacking this condition over data points $x$ gives a linear system $M\,\mathrm{vec}(L) = 0$; the *nullspace* of $M$ (via singular value decomposition, SVD) is the discovered symmetry algebra, and a spectral gap in the singular values separates genuine symmetry directions (near-zero) from non-symmetries. This is implemented in `core/symmetry_discovery.py`. It is a clean linear-algebraic object—the discrete analogue of the width dual certificate—and it succeeds where softer approaches fail precisely because it reads symmetry off a *nullspace* rather than a noisy gradient signal.

The method was validated to recover: 2D rotation ($|\cos| = 1.000$ against the true generator, with a large spectral gap), 3D axis rotation, and—crucially, since the literature flags non-compact groups as delicate—*scaling* (the dilation generator $L = I$, $|\cos| = 0.982$) and the *Lorentz boost* in 1+1D ($|\cos| = 1.000$ against $[[0, 1], [1, 0]]$). Non-compactness poses no difficulty here because the infinitesimal nullspace condition is *local*: it never integrates along the (unbounded) group orbit, so compact and non-compact linear generators are found identically. One honest caveat: the detector reads symmetry off a *trained* reference model's gradients, so it requires a good fit—unlike PCA, which reads raw data covariance, this "preprocessing" contains a model-fitting step.

**Affine extension: translation.** The pure linear ansatz $\dot{x} = Lx$ cannot see *translation* ($\dot{x} = c$, constant), which is the canonical CNN symmetry and non-compact. The fix is the affine ansatz $\dot{x} = Lx + c$, extending the operator with $n$ translation columns (the coefficient of $c_i$ is simply $g_i = \partial_i f$). Discovered generators are classified as translation, linear, or mixed by whether $c$ or $L$ dominates. A subtlety worth recording: unlike the linear detector, the affine operator must *not* be row-normalized, as that compresses the spectral gap for translation generators (whose rows scale differently) and hides the nullspace; the raw operator gives a clean gap. Validated to recover the translation direction $(1, 1, 1)$ on a translation-invariant target while still finding rotations correctly and rejecting generic controls.

### 6.2 Discrete symmetry discovery via equivariance testing

Discrete groups (reflections, permutations, finite rotations) have *no* generator, so the nullspace method is blind to them. The alternative primitive is direct *equivariance testing*: a candidate transformation $g$ (with $g^k = I$) is a symmetry of $f$ iff $f(gx) = f(x)$, measured by the normalized error $E(g) = \mathrm{mean}_x|f(gx) - f(x)|/\mathrm{mean}_x|f(x)|$. This is implemented in `core/discrete_symmetry.py` and covers three families:

- $\mathbb{Z}_2$ (reflections, parity, coordinate swaps): validated to recover the exact involution on even, radial, and swap-symmetric functions, and to report *nothing* on a non-symmetric control. A radial $f(\|x\|^2)$ correctly yields all seven $\mathbb{Z}_2$ involutions of the hyperoctahedral group.

- $S_n$ **permutation subgroups**: since $S_n$ is generated by transpositions, we test the $O(n^2)$ transpositions rather than all $n!$ permutations, then recover the permutable *blocks* as connected components (union-find) of the graph whose edges are discovered transpositions. Validated to recover the exact Young subgroup—$S_3 \times S_2$ on a function symmetric in $\{x_0, x_1, x_2\}$ and separately $\{x_3, x_4\}$; full $S_5$ on a fully symmetric target; trivial on a distinct-weights control.
- $C_n/D_n$ **point groups** (finite rotations and dihedral reflections): the largest $n$ whose full $C_n$ holds, promoted to $D_n$ if a reflection is also present. A robustness subtlety was essential: threshold noise admits spurious isolated orders, so we require *divisor consistency*—$C_n$ is accepted only if every $C_d$ with $d \mid n$ also passes, the subgroup lattice a genuine group must have—and a tightened tolerance separating genuine ($\approx 0.02$) from false-positive ($\approx 0.10$) errors. Validated to recover $C_4$ (reflection-broken), $D_6$, and to saturate at the largest tested order for continuous rotation.

**Detection order is not commutative.** The pipeline runs *continuous first, then discrete*, for a structural reason: any Lie group factors as $G = G^0 \rtimes \pi_0(G)$ (connected part semidirect the discrete component group), so discovering and quotienting the continuous $G^0$ *shrinks* the residual discrete search to $\pi_0(G)$. If the continuous pass is empty, the symmetry is purely discrete and one proceeds directly to the finite-group tests. Running discrete first would flag a swarm of reflections and finite rotations that are merely shadows of an undiscovered continuous symmetry.

### 6.3 Enforcing a discovered symmetry: the commutant equivariant layer

Discovery is only useful if the symmetry can be *imposed*. A linear map $W$ is equivariant to $\exp(\theta L)$ iff it commutes with the generator, $[W, L] = 0$; the set of such $W$ is the *commutant*, a linear subspace found as the nullspace of $W \mapsto LW - WL$. Parameterizing the weight directly in a commutant basis (`core/equivariant_layer.py`) yields a layer that is equivariant *by construction*, and—crucially—*dimension-agnostic*: the same construction gives commutant dimension 2 for a 2D rotation, 3 for a 3D axis rotation, 8 for a 4D double-rotation, all equivariant to machine precision ($\sim 10^{-7}$) with no special-casing. An architectural lesson recorded during validation: equivariance is not invariance. An equivariant layer followed by a plain readout gives *chance* accuracy on an invariant target, because its features rotate with the input; classifying an invariant requires an *invariant* readout (e.g. per-channel squared norms). With that correction the mechanism is correct, though on low-dimensional or easy synthetic tasks the inductive bias provides no measurable *selection* signal—the advantage appears only when the symmetry removes many genuine degrees of freedom (§6.5).

### 6.4 Reduction hierarchy: symmetry-quotient versus principal component analysis (PCA) and the information bottleneck (IB)

The symmetry stage is a form of dimensionality reduction, which invites comparison with PCA. The comparison is sharp and was made empirically on QM7 (rotated-test mean absolute error, MAE; matched downstream head): PCA on raw coordinates is the *worst* reduction, worse than no reduction at all, because PCA removes low-*variance* directions but a rotation *carries* variance—so PCA preserves the nuisance it should remove. A learned information-bottleneck representation does better (it has the right objective, task-relevance) but on scarce data cannot *discover* the rotational nuisance. The symmetry-quotient wins decisively, roughly $2\times$ better than raw and $3\times$ better than PCA, because it removes exactly the SO(3) orbit *losslessly*.

This is not a coincidence but an identity. The IB objective $-I(Y;T) + \beta\, I(X;T)$ is the same variational free energy $\mathcal{F} = R + \frac{1}{\beta}\mathrm{KL}$ (Kullback–Leibler divergence) that organizes the analytical notes: $I(X;T)$ is a description-length (rate) term, $-I(Y;T)$ is the fit, and $\beta$ is the same inverse-temperature/MDL knob. A minimal-and-sufficient representation is provably invariant to all nuisance, so *symmetry-quotienting is the tractable, exact special case of the information bottleneck*—the case where physics hands you the nuisance group, so the minimal sufficient statistic is computable by linear algebra (the commutant/nullspace) rather than estimated by mutual information. An IB-frontier experiment made this concrete: sweeping the bottleneck dimension on the composition-residualized (geometry-dependent) QM7 energy, the symmetry-invariant representation reaches near-optimal accuracy at bottleneck dimension $k$=1 and is flat thereafter (intrinsic task dimension $\approx 1$), whereas raw coordinates remain uncompressible at every $k$—removing the symmetry orbit *collapses* the IB dimension. The two reductions measure the same redundancy.

### 6.5 Real-data validation on QM7

The pipeline was validated end-to-end on QM7 (7165 molecules, atomization-energy regression). The *raw 3D coordinates* are used (`core/qm7.py` loads the arrays $R$ and $Z$), not the Coulomb-matrix representation, which is already rotation-invariant by construction and would pre-bake the symmetry away. Test sets are randomly rotated (the true invariance test). Solo generalization confirms the SO(3)-invariant representation beats a plain coordinate model at every scale (e.g. 42.9 vs 52.8 kcal/mol at $n$=800), with the advantage largest at scarce data.

The metaoptimizer's $\alpha$-selection between an invariant and an unconstrained branch (`core/equivariant_supergraph.py`) is the mechanism by which the symmetry becomes an architectural *decision* rather than a hard preprocessing choice—the appropriate treatment for *approximate* symmetries. Two honest findings: (i) on the small 450-molecule subset the DARTS softmax-mixing selection *failed* (the higher-capacity dense branch dominates the shared objective via co-adaptation), but a solo-comparison selection recovered the invariant branch robustly (4.5× validation-loss gap); (ii) on the full dataset the mixing selection *succeeded* ($\alpha_{\text{inv}} = 0.61$ at $n$=800), showing the earlier failure was a small-data artifact, not fundamental. A real-world caveat is recorded plainly: rotation augmentation of a plain network is a strong simple baseline on QM7, and the mean-pooled invariant descriptor used here underfits relative to state-of-the-art message-passing models; the contribution is the validated *mechanism and selection*, not an absolute accuracy record.

### 6.6 The robust pipeline front-end and its false-positive guards

The cascade is assembled in `core/symmetry_pipeline.py` as `discover_and_reduce`, which runs continuous-then-discrete detection, applies a *re-cascade* (after finding a continuous rotation, discrete detection re-runs on the continuous-invariant features so only genuine residual $\pi_0(G)$ survives, not shadows of the continuous group), and returns a `reduce_fn` implementing the accepted quotient. The subsumption is thereby removed rather than merely flagged: a rotation on $(x_0, x_1)$ plus an independent $\mathbb{Z}_2$ on $x_2$ correctly yields only the $x_2$ reflection.

The central practical lesson—anticipated in principle and confirmed in practice—is that symmetry detection, clean in a pure-mathematical setting, is beset by *false positives* on real, noisy, finite data. Four guards are implemented and were shown necessary: *(G1) refit consensus* (accept a symmetry only if found across a supermajority of independently refit reference models); *(G2) gap/margin* (require a clear spectral gap for continuous, and a clear margin below threshold for discrete); *(G3) divisor/subgroup consistency* (a genuine finite group has its whole subgroup lattice); and *(G4) a null-model test* (the discovered count must exceed a shuffled-label null that destroys real symmetry but preserves smoothness).

A specific, instructive failure exposed the limits of these guards. Run naively on UCR *time-series* data, the discrete detector reported absurd symmetries—a full $S_{23}$ permutation of all time points, plus a hundred $\mathbb{Z}_2$ swaps. The cause is structural, not a mere threshold artifact: on smooth, autocorrelated ordered data, *adjacent coordinates are nearly interchangeable*, so local smoothness masquerades as permutation invariance, and the false positives are systematic enough to survive G1–G3. Equivariance testing simply cannot distinguish "$f$ is smooth in these coordinates" from "$f$ has a group symmetry in these coordinates" when the coordinates are ordered and correlated. The resolution is a `coordinate_structure` flag that encodes the prior every practitioner already has: *ordered* coordinates (time series, sequences, images) disable swap/permutation detection; *exchangeable* coordinates (atoms, point clouds) enable it, gated by the null test. This is recorded as a genuine limitation of the method rather than engineered around.

## 6.7 Head-to-head: does the symmetry front-end help?

The verdict from a controlled comparison (fixed downstream model, symmetry-preprocessed versus raw representation) is twofold and honest. On UCR time series, correctly flagged as *ordered*, the preprocessing is a clean *no-op*—dimension and accuracy unchanged (ItalyPowerDemand 0.973 → 0.973; GunPoint 1.000 → 1.000)—which is the right outcome, since these datasets carry no exploitable coordinate symmetry and the guards prevent false-positive-driven damage. On QM7, where the SO(3) symmetry is genuine and geometric, the same preprocessing delivers a large improvement on rotated-test MAE: 124.4 → 76.3 kcal/mol at $n$=200 (+48.1) and 75.3 → 45.7 at $n$=800 (+29.6), strongest in the scarce-data regime where a symmetry's sample-efficiency advantage is largest.

The two results together are the intended conclusion: symmetry-based preprocessing is a high-value stage precisely when a genuine coordinate symmetry exists, and a safe no-op otherwise—but only with the `coordinate_structure` guard and null test, without which the naive detector is actively harmful on ordered data. This mirrors, at the level of representation, the same conservation law seen throughout the framework: each step toward generality (linear symmetry → discrete symmetry → full information bottleneck) trades a clean, tractable structure for a harder search, and the honest engineering discipline is to use the cleanest tool the data's structure actually admits.

## 6.8 From detection to generation: the autonomous symmetry-to-contract pipeline

The front-end of Section 6.7 discovers *that* a symmetry is present. A second arc, built on the same fit-quality-controlled principle, closes the loop from discovery to a working equivariant architecture, and then extends the group vocabulary far beyond rotations.

**Contract = arch($G$).** Each of the eight contracts is the natural equivariant architecture for a symmetry group. The container type (sequence, grid, set, graph) is a crude proxy for the group; the sharper signal is a *fit-quality-controlled invariance test*. For a candidate group $G$ with metric $g$, we form $G$-invariant features of each datum—inner products $\langle p_i, p_j \rangle_g$ under $g$—and ask whether they explain the target as well as the raw, frame-sensitive coordinates. A Lorentz-invariant target (jet mass) is fit by Minkowski invariants but not Euclidean ones, and conversely; the metric that fits *is* the group.

**Generating the contract (equivariant MLP, EMLP [8]).** Given generators $\{A_k\}$ of $G$, an equivariant linear map $W$ solves the constraint $\rho_{\text{out}}(A_k)\, W - W\, \rho_{\text{in}}(A_k) = 0$; the solution space is the null space of the stacked constraint (Finzi–Welling–Wilson). A crucial subtlety surfaced in validation: pure equivariant-linear-plus-gate networks cannot form the invariant $v \otimes v \to$ scalar, so a metric bilinear layer is required to produce invariants, and the network must map each node to hidden equivariant *vectors*, pool those across the set, and *then* contract to invariants—otherwise cross-node invariants such as the jet mass $\langle \sum p, \sum p \rangle$ are lost. On real JetNet data the generated Lorentz contract reaches $R^2 = 1.000$ on jet invariant mass, versus 0.001 for a plain set network.

**Autonomous group detection.** The loop is closed by choosing the group itself. For each candidate in a menu (Euclidean $\mathrm{SO}(d)$, similarity $\mathrm{Sim}(d) = \mathbb{R}^+ \times \mathrm{SO}(d)$, Lorentz $\mathrm{O}(1, d-1)$), we score a high-resolution, permutation-invariant descriptor—the top-$k$ eigenvalues of the Gram matrix $G_{ij} = \langle p_i, p_j \rangle_g$ together with quantiles of its off-diagonal and diagonal distributions—and select by a *stability* criterion: fit the target on a train split, then score on the test split *and* on a $G$-transformed test split. Because every $\langle p_i, p_j \rangle_g$ is exactly invariant under any transformation preserving $g$, the correct group's fit is stable, whereas a wrong metric collapses (the Euclidean descriptor's fit dropped from 0.42 to $-1.03$ under a Lorentz boost). This stability test is the honest selector: a more expressive descriptor can otherwise overfit frame-dependent signal and choose the wrong metric on in-sample $R^2$ alone. Occam priors resolve ties toward the canonical Euclidean group unless a larger or pseudo-Euclidean group fits strictly better.

**Discovering the metric: arbitrary $\mathrm{O}(p, q)$.** The menu hardcodes only two metrics. The invariant metric of a group is itself discoverable: a generator $A$ preserves $g$ iff $A^\top g + gA = 0$, so given generators the metric is the symmetric $g$ in the null space of the stacked constraint—the same null-space machinery as the equivariant layer, with $g$ as the unknown. The eigenvalue signature $(p, q)$ of $g$, up to an overall sign gauge, names $\mathrm{O}(p, q)$. This is exact on clean generators for every signature tested ($\mathrm{SO}(d) \to \mathrm{O}(d)$; $\mathrm{O}(1, 3)$; the split-signature $\mathrm{O}(2, 2)$; $\mathrm{O}(2, 3)$), and recovers the correct signature from noisy data-discovered generators (the Minkowski metric to $\sim 3\%$, matching LieGAN). Where generator discovery from pooled data is ill-conditioned, a direct route is both cheaper and exact: when the target is a metric norm $y = s^\top g\, s = \langle g, ss^\top \rangle$, it is *linear* in the symmetric matrix $g$, recovered by ordinary least squares. On real JetNet this recovers the exact Minkowski metric $\mathrm{diag}(1, -1, -1, -1)$ at $R^2 = 1.0000$, and it abstains honestly (low fit) when the target is not a metric norm.

**Beyond the metric family.** Three further classes, each with a distinct invariant structure, are detected by a cheap linear regression against that structure with the same fit-quality gate. Complex unitary $\mathrm{U}(n)/\mathrm{SU}(n)$: on $\mathbb{C}^n \cong \mathbb{R}^{2n}$ with complex structure $J$ ($J^2 = -I$), a unitary target preserves $|z|^2$ and is invariant under the phase flow $e^{\theta J}$; the non-vacuous test is the drift of $\langle s, s \rangle_g$ under $e^{\theta J}$ (zero for $|z|^2$, large for a target that splits real and imaginary parts). Symplectic $\mathrm{Sp}(2n)$: preserves a skew form $\omega(u, v) = u^\top \Omega v$, invisible to a symmetric metric, recovered by an antisymmetric *pairwise* regression (skew, nondegenerate, imaginary eigenvalue pairs). Special linear $\mathrm{SL}(n)$: preserves volume, detected by an alternating (Levi-Civita) regression over $n$-vector frames whose coefficients match the permutation signs. Conformal $\mathrm{Conf}(d) = \mathrm{O}(d+1, 1)$: the null-cone lift $x \mapsto (1, x, |x|^2)$ linearizes the conformal action, so the metric regression on lifted pairwise products recovers the $(d+1, 1)$ light-cone signature.

**An autonomous dispatcher.** A single entry point tries every *applicable* route on the *actual* target—never a fabricated one—with applicability read from the data structure (vector pairs for symplectic, frames for special-linear). It resolves overlaps by Occam in the symmetry direction, upgrading a Euclidean detection to $\mathrm{U}(n)$ only when the phase-drift test passes, and inherits the abstention discipline of each detector: if no honest gate passes, it returns none. On real data the dispatcher recovers each detectable class from a physically meaningful target: Lorentz $\mathrm{O}(1,3)$ from JetNet jet mass; Euclidean $\mathrm{O}(3)$ from QM7 and rMD17 molecular geometry; the azimuthal $\mathrm{U}(1)$ of the JetNet transverse plane; the phase-space $\mathrm{Sp}(6)$ of rMD17 position–force pairs; and $\mathrm{SL}(3)$ from the signed volume of QM7 atomic frames—each at $R^2 = 1.0$—while abstaining on non-invariant real targets such as the atomization energy. The discipline throughout is the same conservation law seen elsewhere in the framework: use the cleanest algebraic structure the data actually admits, and decline rather than fabricate when it admits none.

# 7 Metaoptimizing architecture size: width and depth as decisions

The selections of §11.1–§11.4 fix the hidden width at 64 and the depth at 1. This is a real limitation relative to the project's stated aim—a *minimal representation* in which the number of neurons and layers is decided by the metaoptimality criterion, not supplied as a hyperparameter. The width/depth-sparsity machinery (the total-variation width certificate and the priced-depth marginal-value rule) existed in the package but had not been wired into the schema pipeline; the selections above therefore optimized only the primitive *type*. This section reports the integration that closes that gap, so that width and depth become outputs of the criterion.

## 7.1 The marginal-value criterion and the fit–complexity frontier

Capacity is priced. Following the analytical action $\mathcal{J} = R + \mu\cdot(\text{complexity})$, a unit of added capacity (a block of neurons, or a layer) is charged a price $\mu$, and the metaoptimal size is the point at which the marginal reduction in validation loss per unit of added capacity falls to $\mu$. Sweeping $\mu$ traces the fit–complexity Pareto frontier: at low price the criterion buys accuracy with capacity; at high price it buys compactness with accuracy. There is no single "correct" architecture size, only the frontier. Two representative frontiers, measured under the honest bilevel protocol (weights trained on a train partition, the architecture and its marginal curves selected on held-out validation, test accuracy on the official test split), appear in Table 5.

The frontier is genuinely data-dependent. ItalyPowerDemand justifies depth 2 throughout (its second layer measurably reduces validation loss), whereas GunPoint mostly stays at depth 1; and the selected primitive shifts from expensive (`lstm`, `attention`) at low price toward cheap (`plain`, `conv`) at high price. This is the minimal-representation idea realized: width and depth are decisions of the criterion, and $\mu$ is the single knob that places the solution on the frontier.

## 7.2 Depth is usually shallow, but the ceiling is not hand-set

A dedicated probe asked whether any task justifies deeper networks and, if so, whether the selector predicts them. Two findings, one negative and one positive. *Negative:* on compact sequence and tabular tasks depth rarely pays. A multi-hop "pointer-chasing" recall task, which structurally requires composing $K$ sequential lookups, was posed at $K = 3$ and its depth curve measured; validation loss was lowest at depth 1 and the marginal-value criterion selected $L^\star = 1$, the additional layers selecting the parameter-free `norm` stabilizer rather than a compositional lookup. The implemented primitives *compose shallowly* at this scale, and a synthetic label *generated* by a depth-$D$ composition does not create a depth-$D$ *learning* requirement—the composed input–output map is typically smooth and a shallow net fits it (a depth-6-generated label is saturated by a single layer).

The genuinely depth-requiring tasks (parity, deep algorithmic composition) are the ones gradient descent cannot optimize at any depth, so their depth benefit never appears as a measurable marginal—the depth–learnability tension, not a selector defect.

*Positive:* where a real, *learnable* depth benefit exists, the selector does climb to it, and the depth ceiling is not a hand-set constant. `select_architecture_by_area` starts from a small depth grid and, if the best score sits on the grid boundary, extends depth one layer at a time—accepting a deeper column only when it beats the incumbent by more than the across-seed noise floor—up to a generous cap. On ESOL solubility (a graph task where message-passing depth is the physical receptive field), the boundary-extension loop reaches beyond the initial depth-3 grid and selects $L^\star = 4$ (validation $R^2$ $0.70 \to 0.77$ at depth 4), continuing to depth 5 only where the gain clears the noise floor and stopping there; on the large-molecule subset the area rule instead selects a narrow-deep $w16/L3$ over wider shallow options at matched accuracy. A methodological caveat governs this: the significance gate must use at least two standard errors—a one-standard-error test manufactures spurious depth (it assigned a linearly separable control depth 3; at two standard errors the control correctly returns depth 1), the same seed-noise lesson as the depth-scaling study (§13). So the practical default sweeps $L \in \{1, 2, 3\}$ with boundary extension enabled, which is conservative on the shallow majority yet reaches depth 4 on ESOL rather than being capped there (`studies/deep_network_depth_study.py`).

### 7.3 Compact and accurate: a differentiable complexity penalty

The marginal-value stopping rule has a weakness: its high-accuracy end is capped, because the per-width marginal can fall below $\mu$ before the sweep reaches the width that would maximize accuracy. A related concern—that the sweep's fixed upper width is an arbitrary bound that can be *binding* rather than a true optimum—is now addressed directly: when the priced rule selects the largest candidate because the marginal never fell below the price (the ceiling is binding, as on the still-climbing rMD17 width curve), the sweep doubles the ceiling and re-measures until the rule makes a genuine interior stop or a saturation floor halts it, so the chosen width is an output of the priced criterion rather than of where the grid happened to end. The resource-aware differentiable neural-architecture-search (NAS) literature supplies a better instrument for the marginal-threshold issue, and one that sits more naturally in the variational framing: fold the complexity cost directly into the differentiable architecture objective rather than applying it as a post-hoc threshold. Each primitive $i$ is charged its actual (normalized) parameter cost $c_i$, and the architecture loss gains the term $\mu \sum_i \mathrm{softmax}(\boldsymbol{\alpha})_i \, c_i$, so gradient descent on $\boldsymbol{\alpha}$ is pulled toward cheap primitives *during* search, in proportion to $\mu$—exactly the action $\mathcal{J} = R + \mu \cdot (\text{complexity})$ realized as a single differentiable objective. An entropy-sharpening term mitigates the discretization gap, and width is chosen by accuracy-first compaction (the smallest width within a tolerance of the best validation accuracy), which preserves the high-accuracy end the marginal rule loses. Table 6 reports this method across the real datasets, over the full nine-primitive vocabulary. (The fixed-width selections of §11.2–§11.4 were run over the earlier eight-primitive vocabulary, before `linssm` was added; this is why, for example, BasicMotions selects `lstm` there but `linssm` here—the newer primitive is available only in the nine-way runs. The size-metaoptimization protocol used here is the benchmark-of-record for newly-added primitives going forward.)

Two honest observations qualify these numbers. First, the penalized accuracies run below the fixed-width-64 references of Table 12 because two effects compound: the compaction itself, and the switch to a bilevel (held-out-validation) selection protocol, which is more defensible but costs some accuracy relative to joint training on all data. The comparison to keep in view is not penalized-vs-fixed but the shape of the tradeoff: BasicMotions loses nothing at a smaller size, and the others buy real compaction (up to an order of magnitude fewer parameters on ACSF1) at a measured accuracy cost. Second, under strong complexity pressure the $\boldsymbol{\alpha}$ distribution stays comparatively flat, because several cheap primitives (`dense`, `norm`, `spectral`) have both similar cost and similar, mediocre fit; the argmax among near-ties is then somewhat arbitrary. Under heavy compaction the meaningful signal is the size reduction, not the identity of the selected cheap primitive.

### 7.4 Minimal size on the synthetic tasks

The synthetic tasks make the minimal-representation idea especially vivid, because their ground-truth structure fixes how much capacity is genuinely required. Associative recall is solved perfectly (accuracy 1.000) already at width 4—the smallest tried—selecting `attention` throughout, for a deployed network of only 3,714 parameters: once the correct routing primitive is identified, the task needs almost no capacity. Copy is different: at width 8–16 it fails (the criterion selects a degenerate cheap primitive at chance-level accuracy), and a genuine minimal width near 32 is required before a non-trivial solution (`attention`, accuracy $\approx 0.60$) appears. Different tasks therefore have genuinely different minimal capacities—a few thousand parameters for recall, an order of magnitude more for copy—and the criterion recovers this rather than imposing a uniform size.

### 7.5 Generalized area: variable width-per-layer, emergent depth, and a calibrated price

The width/depth selectors above choose a *uniform* width and a depth. The generalized-area selector (`core/variable_width_area.py`) prices the single quantity $R + \lambda \sum_\ell m_\ell$—the total neuron count with the per-layer structure kept—so that both the per-layer width and the depth are outputs of one functional (depth is emergent, $L = \#\{\ell : m_\ell > 0\}$, a closed layer costing zero neurons). Three implementation points made it usable in practice.

*A deterministic gate for a low-noise price path.* A stochastic hard-concrete ($L_0$) gate makes the selected widths vary across seeds because the neuron count is a Monte-Carlo estimate; the price path is then too noisy to read. Replacing it with a deterministic gate $g = \sigma(\text{logit}/\tau)$ whose temperature $\tau$ is annealed toward zero makes the reported area a deterministic function of the parameters. On ESOL the across-seed standard deviation of the held-out $R^2$ dropped from $\sim 0.2$ (stochastic) to $\sim 0.01$ in the accuracy-retaining regime, and the width$\to$area path became monotone.

*A certificate-calibrated price.* Rather than sweeping $\lambda$ on an arbitrary grid, the dual certificate sets its scale: `certificate_lambda_scale`$(X, y) = \|A^* y\|_\infty / n$ over a random-neuron dictionary is the peak residual–atom correlation, the price at which the first neuron stops being worth adding (ESOL: $\lambda_0^\star \approx 0.81$). `area_price_path` then sweeps small fractions of this scale, so the range lands where the price actually arbitrates neurons. This fixes the *scale* of $\lambda$ per dataset from a measured quantity; it is not a universal scalar (it scales with the residual, and the deep gated network's effective per-neuron price is a fraction of the single-atom value), so the final choice is still made on the accuracy/area frontier.

*Integration as a size mode.* `AllGraph.fit(select_size=...)` now takes a mode rather than a Boolean: `"sequential"` (the priced width-then-depth selector), `"area"` (joint uniform width$\times$depth minimization), and `"variable"` (the generalized-area selector). The variable mode featurizes the routed data to a fixed vector—permutation-invariant mean$\|$max pooling for relational data, flattening or pooled summaries for dense sequence/grid data—runs the certificate-calibrated minimization, and writes the emergent depth and the largest kept width back to `self.depth`/`self.width`, after which the routed *contract* schema is trained at that size. The variable-width analysis is thus a size *recommender*; the deployed model remains the contract's own schema with its primitives. Because the featurization is to a vector, the variable mode applies to sequence and grid data as well as relational data, whereas the sequential/area selectors remain relational-only (they train candidate contracts directly).

*A principled depth ceiling.* The variable selector does not cap depth at a hand-set constant. It starts at a small maximum depth and extends it by a significance-gated rule—grow while the ceiling is binding (the net uses all its layers) *and* one more layer improves the probe accuracy by more than the across-seed noise, checked from the first extension—so a task that already fits stops immediately. On real data this differentiates correctly: a linear task stops at the base depth, ECG5000 extends, ESOL and GunPoint stop at depth 3. In a mid-preset run with the variable mode across seven relational datasets it produced tapering "funnel" width profiles (e.g. ESOL $[45.5, 38, 32.5, 26]$) and improved accuracy on five of seven at matched budget, the two regressions (Tox21, rMD17) being an over-capacity small-signal task and an under-trained one respectively—consistent with the rule that deeper helps only sometimes.

*The one contract outside the size stage: spectral modes.* The width/depth selectors above address cell width and layer count, but the operator (Fourier neural operator) contract has no cell width to select—its capacity is set by the number of retained Fourier *modes*, and the mode budget was fixed by a heuristic (and, as an audit found, hardcoded per spectral primitive, so the nominal budget argument did not even reach the spectral convolution). The mode count is nonetheless a degrees-of-freedom knob—a spectral convolution stores a learnable complex weight per retained mode, so its parameter count grows linearly in the budget per axis—and it is now priced by the same marginal-value rule as depth (`machinery/spectral_selection.py`). The selector measures validation loss against a ladder of mode budgets and retains modes while the per-mode loss reduction exceeds $\mu$ times the added spectral code length, the latter taken as the logarithm of the retained-mode parameter count so that a single price governs modes, width, and depth on a common footing. The marginal value of modes has exactly the structure the rule needs: a band-limited target (most PDE solution operators are smooth, hence low-wavenumber) gains sharply from the first few modes and then nothing, with the marginal turning negative once excess modes begin to fit noise. On the one-dimensional Burgers operator the selected budget is a small fraction of the heuristic at unchanged field accuracy; on two-dimensional Darcy flow, where the spectral weight count grows *quadratically* in the budget, the priced selection cuts the parameter count several-fold while slightly improving accuracy. This brings the operator contract into the physicist's ordering: the mode budget joins width and depth as a measured, priced quantity rather than a fixed number.

### 7.6 Closing three fixed-hyperparameter gaps: kernel size, angular order, nonlinear symmetry

Three standalone "derive-the-hyperparameter" modules were wired into the degrees-of-freedom stage as opt-in flags (default off; the default path is unchanged and regression-checked), each turning a hand-fixed hyperparameter into an output of the same measured/priced principle the size selectors use.

*Kernel size from the correlation length* (`kernel_from_xi`). The spatial schema already carries larger-kernel primitives ($k = 5, 7$) that compete by selection, but the default menu omits them. `correlation_length` measures the spatial correlation length $\xi$ and, when the receptive field warrants it, *augments the menu* with the larger kernels; the selector then chooses. This is menu augmentation, not a hardcoded-$k$ override, and it respects that the schema selects among what it is given. On smooth fields ($\xi \approx 3.6$) it added $k5/k7$ and the selector chose $k7$; on noisy fields it stayed lean; through the runner MNIST rose from 0.760 to 0.833.

*Angular order from the data* (`angular_from_data`). `angular_resolution` applies the priced marginal-value rule to angular order: order $\ell$ is kept only while the target variance it explains, that lower orders cannot, exceeds a price. In the current $\ell \leq 1$ equivariant builder this decides whether the $\ell = 1$ vector channels are warranted (radial target $\Rightarrow c_1 = 0$, a leaner scalars-only net). On QM7-equiv it selected $\ell = 1$, correctly keeping the vector channels molecular energies require.

*Nonlinear symmetry as an opt-in diagnostic* (`nonlinear_symmetry_fallback`). The linear Lie-derivative detector sees only symmetries acting linearly in the given coordinates. The nonlinear (LaLiGAN [29]) detector finds symmetries linear only after a learned coordinate change—a strict superset—but it trains an autoencoder and is bounded by reconstruction quality. As a diagnostic it *reports* a nonlinear symmetry without switching the contract. A second, deliberately separate opt-in (`deploy_nonlinear_contract`) now closes the loop for the nonlinear case: when the joint symmetry-regularized autoencoder *confirms* a latent symmetry (under a null-baseline guard and a non-degeneracy check), the package builds and deploys the *latent-equivariant contract* $x \mapsto \phi(x) \mapsto \mathrm{EMLP}(L_z) \mapsto y$—the encoder $\phi$ that discovered the symmetry composed with an equivariant network built (by the same EMLP machinery as the linear generated contract) from the linear latent generators $L_z$. On a controlled task with a genuine nonlinear symmetry this contract is exactly invariant in the latent coordinates and its advantage over a plain network grows as data shrinks (the low-data inductive-bias gain). Two honest limits bound it: the joint discovery is conservative and on real data typically does *not* confirm (so the contract falls back to the normal route, never degrading it), and the benefit is contingent on the learned chart's fidelity—a poor chart can underperform the plain network. It is null-guarded and off by default: the safe action for a superset capability, now extended from diagnosis to a gated deployment when—and only when—discovery genuinely confirms.

**A scalable realization of the equivariant head.** Both the linear generated contract and the latent-equivariant contract build their networks with the exact EMLP construction, which realizes each equivariant linear layer by solving for its basis—an SVD of a Kronecker constraint matrix of size $D^2 \times D^2$ for a representation of dimension $D$. That solve is cubic in $D^2$ and, measured in this codebase, explodes from a millisecond at $D = 4$ to seven seconds at $D = 48$: EMLP is exact and fully general but does not scale, which is why in the literature it is largely confined to small synthetic problems. The package therefore also carries a scalable realization (`models/scalable_equivariant.py`) in the Vector-Neurons [7] / G-RepsNet [2] family (Deng et al. 2021; Basu et al. 2024): features are kept as explicit vectors in the group representation and combined only by operations that are equivariant *by construction*—a scalar-weighted mixing of vector channels ($v'_o = \sum_i W_{oi} v_i$ transforms as a vector under any linear action), metric inner products $\langle v_i, v_j \rangle_M$ as invariants (the metric $M$ making them group-correct, identity for $\mathrm{O}(n)$, $\mathrm{diag}(1, -1, -1, -1)$ for the Lorentz group), and a Vector-Neurons gate $v \mapsto \phi(\|v\|_M)\, v$. No constraint is solved, so the cost is linear in the number of channels rather than cubic in the representation dimension. The realization is exactly equivariant (verified to floating-point precision for SO(2), SO(3), and the Lorentz group) and, on a synthetic invariant-regression benchmark, builds twenty-two to eighty-five times faster than EMLP with the gap widening as the representation grows; at the point where EMLP's basis becomes large its optimization degrades and the scalable head is both faster and markedly more accurate. It is offered alongside—not in place of—EMLP, which remains the default and the choice when its full generality (every higher-order invariant) is wanted on a small representation.

**Pricing approximate equivariance.** Real data is often only approximately symmetric, yet the geometric contract as described is a hard choice—strictly equivariant or not. The amount of equivariance can instead be a *selected* quantity, priced by the same objective $J = R + \mu\Omega$

that governs every other structural decision. The model is a residual pathway (Finzi et al. 2021), $f(x) = f_{\mathrm{equiv}}(x) + \rho\, f_{\mathrm{free}}(x)$, with $\rho \geq 0$ the relaxation strength: $\rho = 0$ recovers exact equivariance (the package's exact head, so exactness is inherited), $\rho > 0$ admits controlled symmetry breaking. The key subtlety, emphasized in the approximate-equivariance literature (van der Ouderaa & van der Wilk), is that symmetry is a *constraint* and is therefore never rewarded by a training loss that measures fit: relaxing always lowers training loss, so $\rho$ cannot be chosen by fit and must be paid for. The symmetry-breaking pathway is added structure, and the honest code-length charge is the fraction of the output that actually flows through it, $\Omega(\rho) = \mathbb{E}\|\rho f_{\mathrm{free}}\|^2/\mathbb{E}\|f\|^2 \in [0, 1)$—zero when the free path is unused, growing as breaking is used. Selecting $\rho$ by $\arg\min_\rho R_{\mathrm{val}}(\rho) + \mu\, \Omega(\rho)$ over a small ladder then does the right thing: on exactly-symmetric data the strict model already minimizes held-out risk and $\Omega = 0$, so $\rho = 0$ is chosen (Occam toward symmetry); on symmetry-broken data the strict model's risk is large and the fit gain overrides the $\Omega$ charge, so a positive $\rho$ matched to how broken the data is. On a controlled pendulum the strict model's validation risk is an order of magnitude higher under an added symmetry-breaking term than without it, which is exactly the signal the price responds to, and raising $\mu$ pulls the selection back toward exact equivariance more readily on the near-symmetric case than on the broken one. This prices a single scalar relaxation for one group—not the per-layer, per-group relaxation of the mixed-symmetry literature—and $\Omega$ is a fit-measured code-length proxy rather than a certified evidence term, but it places "how much symmetry" on the same priced, held-out-validated footing as the rest of the selection. Running the full relaxation ladder is four model fits, which is costly on the equivariant contract—the most expensive contract—and wasteful on data whose symmetry is in fact intact. The main equivariant path therefore carries a nearly-free *breaking probe* instead: after the strict fit, a single linear least-squares solve tests whether the residual the equivariant model cannot represent is predictable from a non-invariant feature (the coordinate sum, whose components break the rotation) beyond an invariant one (the radius); a positive signal flags broken symmetry and hence that the priced relaxation is worth running. On real molecular energies, which are genuinely rotation-invariant, the probe reports the symmetry intact and the strict model is kept; on a field-broken variant it flags the breaking, matching the priced ladder's own verdict at a fraction of the cost.

# 8 Primitives and schema realizations

The *dynamics* rung trains the couplings, but first it needs a fixed field content: the set of candidate operations (primitives) the architecture parameter will mix over, and the per-contract schemas that hold them. This section fixes that vocabulary—the six irreducible primitives (§8.1), the four high-value composite primitives added on top of them (§8.2), and the eight concrete schema realizations (§8.3).

## 8.1 The primitive vocabulary: six irreducibles, eleven sequence realizations

The analytical work identified a minimal spanning set of six irreducible primitives—the operations from which the modern architecture zoo is composed and which cannot be composed from one another: (1) affine map, (2) pointwise nonlinearity, (3) multiplicative gating, (4) weight sharing under a group action, (5) content-based routing, and (6) normalization. The sequence contract exposes *eleven* selectable entries, and the discrepancy is not a contradiction but a consequence of how irreducible primitives become useful selectable units. Table 7 reconciles the counts.

Two facts drive the count from six to eight base entries (a ninth, `linssm`, is added below as a linear-recurrence variant). Weight sharing occupies two entries because its internal group parameter (translation vs. time) is discrete and non-relaxable, so the two useful instantiations must be separate candidates. Gating occupies two entries because `gated` and `lstm`, though built from the same irreducibles, have materially different optimization profiles—`lstm` requires chrono-initialization [23] to converge on long-range tasks, a distinction worth exposing to the selector. Pointwise nonlinearity is always-on and never a standalone candidate.

## 8.2 Four high-value primitives

A vocabulary audit against the architecture literature identified four families with no representative in the original registries; each gap was closed by one core and validated on real natural-science data.

**Multi-scale temporal (`dilconv`) and spatial (`atrous`).** The original `conv` is a single small-kernel convolution (the shortfall flagged in the first report). A dilated-convolution core with parallel causal branches at dilations $1, 2, 4, 8$ (the temporal-convolutional-network, TCN [1]/InceptionTime and atrous-spatial-pyramid-pooling, ASPP/DeepLab constructions) gives a multi-scale receptive field while remaining exactly shift-equivariant. On OSULeaf (leaf-outline botany) `dilconv` beat `conv` by 5.8 accuracy points; on MNIST `atrous` beat `conv2d` by 22 points. Both are genuine expressiveness gains, not reparametrizations.

**Multi-aggregator graph messages (`pna`).** Single-aggregator message passing is provably limited: mean aggregation cannot distinguish a node with neighbour multiset $\{1, 3\}$ from one with $\{2, 2\}$. Principal Neighbourhood Aggregation combines mean/max/min/std aggregators with degree-scalers, resolving the ambiguity and, on QM7 energy regression, giving the best clean-solo $R^2$ of all graph primitives (0.572 versus 0.517 for GIN [28] (graph isomorphism network), 0.471 for GCN [14] (graph convolutional network)).

**Scalar–vector equivariant messages (`e_painn`).** The original equivariant core `e_tp` uses full Clebsch–Gordan tensor products [4, 3]. PaiNN's [21] scalar–vector message passing is a distinct, cheaper philosophy: it maintains equivariance with only scalar and vector channels and directional updates via the edge unit vector, never mixing vector components through a non-equivariant operation. Both are exactly SO(3)-equivariant (scalar invariance and vector equivariance at machine precision, $\sim 10^{-7}$), and they win in *different* regimes: on data-scarce QM7 the richer tensor-product coupling of `e_tp` wins ($R^2$ 0.72 vs 0.65), whereas on the molecular-dynamics regime of rMD17 ethanol `e_painn` wins by seven points (0.736 vs 0.664). This dataset-dependence is precisely why both belong in the vocabulary: the metaoptimizer selects the right one per task.

## 8.3 Implemented schemas

A schema runs several primitives in parallel and mixes their outputs by a softmax over the architecture parameter $\boldsymbol{\alpha}$, so that gradient descent on $\boldsymbol{\alpha}$ selects a primitive. Six schema modules were built, in increasing scope, each leaving its validated predecessors untouched. Table 8 inventories them.

### 8.3.1 The interface split and its resolution

The first five schemas partition the primitives by *interface*: recurrent cores consume a sequence and thread state through a per-timestep loop; the parallel and spatial cores consume a whole vector or image in one shot. A consequence, easy to miss, is that a recurrent primitive and a non-recurrent one never competed on the same task—the vocabulary was complete only *across* the framework, never within a single selectable schema.

The schema (`schema.py`) resolves this. Every primitive is given a common sequence-to-sequence contract, `forward_seq`: $(b, T, n_{\text{in}}) \to (b, T, \text{width})$. Recurrent primitives realize it via their timestep loop; `conv` as a causal 1-D convolution over time; `attention` as causal self-attention over timesteps; `dense` and `norm` per timestep; `spectral` by an rFFT over the time axis whose (time-invariant) summary is broadcast back. A per-layer softmax over $\boldsymbol{\alpha}$ mixes the eight sequence outputs, with a shared readout head; a configurable readout (last timestep, for recall/copy; mean over time, for classification) collapses the sequence to a prediction. This is the honest realization of the six-primitive design: a single schema in which the whole vocabulary competes, and from which a concrete architecture is read off as the per-layer arg max of the peak architecture weights.

*Note.* Table 8 details the sequence schema specifically; it is one of *eight* peer schemas—sequence, spatial (2D), volumetric (3D), spatiotemporal (4D), graph, equivariant-graph, set, and operator (neural-operator)—each a self-contained contract with its own primitive registry, all unified behind the single meta-router inventoried in §4.1 (dispatch in §4.2). The six-irreducible analysis of §8.1 covers all of them: the schemas realize the *same* six irreducibles under different group actions and tensor ranks, with a larger count of selectable realizations per contract (Table 2).

# 9 The architecture parameter

With the vocabulary fixed, the architecture parameter $\boldsymbol{\alpha}$ selects among the primitives. Three questions arise, treated in turn: how to select robustly when primitives co-adapt inside the trained mixture (§9.1); how $\boldsymbol{\alpha}$ is *derived* rather than imported, as the outer Gibbs measure of a free energy on the primitive simplex (§9.2); and how the mixture is *priced*, so that a sparse committed selection is preferred to a hedged blend (§9.4).

## 9.1 Robust primitive selection under co-adaptation

Selecting a primitive from a trained supernet is complicated by co-adaptation: because the primitives are trained *together* inside the softmax mixture over $\boldsymbol{\alpha}$, the argmax of the architecture weight reflects “who helps the mixture”, not “who is best alone”. The literature’s robust fix is selection-phase ablation (DARTS-PT): set the architecture weight to a one-hot on each primitive in turn and score its isolated validation loss. This defeats co-adaptation for parameter-free or low-capacity co-adapters (a normalization or skip branch that only fits residuals collapses when isolated), and that case was validated.

**A boundary of the ablation proxy.** The PNA (principal neighbourhood aggregation [6]) case exposed a genuine limitation. Ablation under-rated `pna`: forced solo on its *mixture-trained* weights it scored worst, yet trained from scratch it was best. The reason is structural—a high-capacity core with a large internal projection and an additive residual has weights co-adapted to the mixture context; isolating those weights (which never learned to solve the task alone) is a false negative. The reliable readout is therefore *clean-solo*: train each candidate primitive alone, from scratch, and rank by held-out score. This is more expensive (one training run per primitive) but avoids the false negative when candidates differ in capacity.

**A contract-agnostic protocol.** Testing ablation against clean-solo across contracts showed the pathology is data- and architecture-dependent, not predictable from capacity: it fired on graph/`pna` but *not* on spatial/`atrous` or sequence/`dilconv`, all of which are also high-capacity multi-branch cores. Robustness therefore cannot be a per-case manual judgement; it is packaged as a single entry point `robust_select` that (i) measures the argmax-versus-solo disagreement rate, (ii) takes the fast ablation path when disagreement is low, and (iii) arbitrates with clean-solo only when disagreement is high—paying the expensive comparison exactly when co-adaptation is actually present. On the graph/PNA case it correctly triggers clean-solo arbitration; on spatial/atrous it takes the fast path and never calls clean-solo.

### 9.2 Deriving the architecture parameter

The co-adaptation analysis of §9.1 and the standing seam of §2.3—that $\boldsymbol{\alpha}$ is imported from DARTS rather than derived—turn out to be the same problem, and it admits a single resolution. The companion analytical report derives $\boldsymbol{\alpha}$ as the Gibbs stationary measure of a free energy on the primitive simplex; here we record the implementation (`machinery/gibbs_alpha.py`) and its empirical payoff.

**The derived selector.** Put on the primitive simplex the free energy $\mathcal{F}_\alpha = \langle \boldsymbol{\alpha}, \Psi \rangle - \frac{1}{\beta} H(\boldsymbol{\alpha})$, the same $R + \frac{1}{\beta}$(rate) form used for the neuron measure, with $\Psi_p$ the *solo* marginal risk of primitive $p$. Its stationary measure is the Gibbs law $\boldsymbol{\alpha}_p \propto e^{-\beta \Psi_p}$; its Fisher–Rao gradient flow is the replicator equation; and DARTS softmax-gradient-descent is that flow in a preconditioned metric, sharing its entire critical set (verified symbolically for $P = 2$ as an exact time-rescaling, numerically for general $P$). The knob $\beta$ behaves as $\mu$ does for depth: $\beta \to \infty$ concentrates on $\arg\min_p \Psi_p$ (the hard, single-primitive certificate), finite $\beta$ keeps a hedged mixture, and sweeping $\beta$ traces a fit-versus-commitment frontier (`gibbs_frontier`, `select_beta_by_elbow`). The selector `gibbs_alpha_select` consumes exactly the clean-solo scores of §9.1, so no new training machinery is needed.

**Payoff on the co-adaptation case.** On the QM7 graph task, with the vocabulary {`gcn`, `gin`, `pna`, `norm`}, argmax-DARTS on the trained mixture selects the parameter-free `norm` co-adapter—the *worst* clean-solo primitive—in every seed, while the derived Gibbs-$\boldsymbol{\alpha}$ selects `pna`, the clean-solo best, in every seed and at every temperature $\beta \in \{1, 4, 8, 20, 100\}$. The derivation does not merely rename the softmax; because its energy is a solo quantity it cannot see the mixture co-adaptation that misleads the argmax.

**Integration and deployment.** The selector is wired into the meta-router as `AllGraph.fit(..., select="gibbs")`: after the mixture trains, each layer's primitive is reselected by the derived Gibbs-$\boldsymbol{\alpha}$ over clean-solo energies, and the winner is deployed as a fresh single-primitive network trained on the full data (the mixture weights, being co-adapted, are not reused). Table 9 contrasts the deployed derived architecture with the argmax mixture on the relational contracts, at a smoke budget.

**Historical note (the superseded "bolt-on" framing).** For the record, earlier iterations of this work treated $\boldsymbol{\alpha}$ as a pragmatic import from DARTS with *no* derivation from the variational framework—a "bolt-on." The reasoning was that the continuum machinery (measure lift, dual certificate, path integral, depth RG) is *within*-architecture: it presupposes a fixed computational graph and analyzes training, width, depth, or initialization inside it, whereas $\boldsymbol{\alpha}$ chooses *between* graphs, a discrete selection with no Euler–Lagrange equation. The observation that an entropic free energy on the primitive simplex has softmax as its stationary measure was, at that stage, regarded as an analogy rather than a load-bearing unification. The derivation above (with DARTS as the Fisher–Rao gradient flow and the co-adaptation-free solo marginal risk as the forced energy) makes it load-bearing *within* a contract; what survives of the original seam is only the discrete cross-contract dispatch (§2.3, §4.2), which the interface-incompatibility argument shows genuinely cannot be relaxed. This paragraph replaces the separately-archived legacy note.

### 9.3 One free energy, one form, three temperatures

The derivation above writes the primitive readout as the Gibbs measure of $\mathcal{F}_\alpha = \langle \boldsymbol{\alpha}, \Psi \rangle - \frac{1}{\beta} H(\boldsymbol{\alpha})$. The same free-energy *form*

$$\mathcal{F}(\rho) \;=\; \langle \rho, E \rangle \;-\; \tfrac{1}{\beta}\, H(\rho), \qquad \rho \propto e^{-\beta E} \text{ (interior stationary measure)}, \tag{1}$$

governs *every* priced decision in the package, and each carries an inverse temperature $\beta$. Because the symbol recurs—the primitive readout's $\beta$ (`gibbs_beta`), the WBIC $\beta = 1/\log n$ inside the local-learning-coefficient estimator [15, 27] (the singular free energy $F_n = nL + \lambda \log n$), and the implicit unit temperature of the contract posterior (§4.3)—a careful reader will ask whether these are the same temperature. They are not, and the distinction is load-bearing.

**Three levels of one functional.** Equation (1) is instantiated at three successive coarse-grainings of the model, which differ only in the space the measure lives on, the energy on that space, and hence the natural value and units of $\beta$:

| Level | space | energy $E$ | temperature | pinned by |
|---|---|---|---|---|
| weights $w$ | weight space | $n\,L(w)$ (nats) | $\beta_W = 1/\log n$ | WBIC theory ($\to \lambda$) |
| primitives $\boldsymbol{\alpha}$ | primitive simplex | $\Psi_p$ (solo fit cost) | $\beta_A$ = `gibbs_beta` | readout sharpness (free) |
| contracts $c$ | contract set | $n\hat{L}_c + \Omega_{\text{struct}}(c)$ (nats) | $\beta_C = 1$ | nats-vs-nats bookkeeping |

The derivative read-outs are, respectively, the local learning coefficient $\lambda = \mathrm{d}F_n/\mathrm{d}\log n$ [15]; the derived architecture distribution $\boldsymbol{\alpha}$ (§9.2); and the contract posterior, whose MAP coincides with the priced-$J$ winner (§4.3). These are the same functional at the weight, mixture, and contract scales—not three unrelated uses of a letter.

**Why the temperatures are decoupled (and must not be equated).** It is tempting to "unify" by setting the primitive temperature equal to the WBIC temperature, `gibbs_beta` $:= 1/\log n$. This is wrong. At $n = 1200$, $1/\log n \approx 0.14$ whereas the default `gibbs_beta` $= 8$; equating them drops the readout temperature by $\sim 57\times$ and collapses $\boldsymbol{\alpha}$ to nearly uniform, destroying the selection. The three energies moreover live on different spaces and in different units—$nL$ is $O(n)$ nats on weight space, $\Psi_p$ is an $O(1)$ dimensionless fit metric on the simplex, $n\hat{L}_c + \Omega_c$ is $O(n)$ nats on the contract set—so there is no single partition function $Z_n(\beta)$ whose marginals are all three. Each temperature is pinned by a *different* principle: $\beta_W = 1/\log n$ is theory-fixed (only this value makes $n\beta_W(\mathbb{E}_{\text{post}}[L] - L^\star)$ estimate the real log canonical threshold), $\beta_C = 1$ is unit-fixed (an NLL against a description length, both already in nats, admits no free exchange rate; the optional $\mu_c$ only tempers the prior), and $\beta_A$ is a genuine free sharpness knob. The reconciliation is therefore one free-energy *form*, three coarse-grainings, three deliberately decoupled temperatures—not one temperature.

The package records this as a single potential (`machinery/thermodynamic_potential.py`) whose `assert_temperature_consistency` verifies each $\beta$ sits at its principled value and, in particular, flags the coupling error $\beta_A = \beta_W$; an opt-in `report_thermo` attaches the hierarchy to a fit result. This closes the latent notational double-use that Eq. (1)'s recurrence would otherwise invite, and is the thermodynamic backbone from which the singular complexity [15] and the developmental read-out are read as derivatives of the *same* object.

### 9.4 Pricing the mixture: sparsity-priced $\boldsymbol{\alpha}$

The derivation of §9.2 selects a single primitive. A natural question is whether it extends to a genuine *mixture*—a deployed net that computes $\sum_p \boldsymbol{\alpha}_p f_p$ and can beat any single primitive. The honest answer (developed in the companion analytical report and recorded in the package's derivation notes) is a delimitation: the dense in-layer mixture cannot be *derived* beyond DARTS, because for a live network $\boldsymbol{\alpha}$ and the weights minimise the *same* non-convex loss, so the "derived" $\boldsymbol{\alpha}$ is just entropic mirror descent on that loss—a reparametrisation of DARTS, not a new principle. (The one case with a clean certificate is a *frozen* basis, i.e. post-hoc ensembling, which is ruled out here: a bag of separately trained models violates the single-minimal-net discipline.)

What survives, and fits the framework's logic exactly, is to *price* the mixture rather than derive it. Charge the mixture its description length and let a single price decide how many primitives it keeps.

**The charge is the effective primitive count.** The MDL two-part code for a mixture over $P$ primitives is $\sim k \log P + k \cdot \text{(weight bits)}$ with $k$ the number of active primitives—linear in the support size $\|\boldsymbol{\alpha}\|_0$. Since $\|\cdot\|_0$ is combinatorial, the smooth surrogate is the *inverse participation ratio (IPR)*

$$\mathrm{IPR}(\boldsymbol{\alpha}) = \frac{1}{\sum_p \boldsymbol{\alpha}_p^2} \;\in [1, P], \tag{2}$$

the physics "participation ratio": 1 for a single primitive, $P$ for a uniform mixture. Pricing it means rewarding concentration, $\Omega(\boldsymbol{\alpha}) = -\sum_p \boldsymbol{\alpha}_p^2$, so the architecture objective becomes $\mathcal{J} = R + \mu\,\Omega$—the identical priced form as the width (effective neuron count) and depth (layer count) charges, now with the *effective primitive count* as $\Omega$. It plugs into the same price-selection selectors (budget, elbow, validation) verbatim. Entropy is deliberately *not* used: it prices spread rather than count, and is already the $1/\beta$ soft-selection term; the IPR literally counts branches. Implemented in `machinery/sparsity_priced_alpha.py`; a differentiable `sparsity_price(`$\boldsymbol{\alpha}$`)` $= -\sum_p \boldsymbol{\alpha}_p^2$ is added per cell to the live training loss, and `AllGraph.fit(...,select="sparse")` deploys the resulting compact mixture (never forced to collapse).

**What the price does on real data.** Table 10 sweeps $\mu$ on the graph contract. The behaviour is exactly "let the frontier decide": one price, and each dataset lands where its own accuracy-versus-IPR frontier dictates.

The upshot is that "extend $\boldsymbol{\alpha}$ to mixtures" is answered in the framework's own idiom: not by a richer derivation (which collapses to DARTS) but by a price with a certificate-flavoured reading—a primitive stays in the mixture only while its marginal risk reduction exceeds its IPR price $\mu$, the same conditional-gradient logic as the width and depth rungs, one level out on the primitive dictionary.

## 10 Interpretability as a corollary, and redundancy reduction

Because the description-length action selects a sparse, discrete architecture, the selected object is itself the explanation—there is no separate surrogate to interrogate. Three faithful-by-construction tiers (`core/interpretability.py`, `feature_attribution.py`, `symbolic_readout.py`) read out the selection at three levels, and each was validated on real data with its failure mode made explicit.

*Tier 1, architecture as explanation.* `explain/AllGraph.explain` reports which primitive each cell chose, from the clean-solo Gibbs energies (falling back to the per-cell $\alpha$ peak so the report matches the deployed architecture), pairing the argmax with the participation ratio and surfacing ties rather than breaking them silently. On ECG5000 it reports an `lstm/dilconv` tie; on GunPoint, where the data is too small to separate primitives, it faithfully reports all primitives tied—diagnosing the known under-training rather than inventing a winner.

*Tier 2, feature attribution.* A hard-concrete $L_0$ feature gate exposes which invariant coordinates carry the signal. The key finding is a boundary, not a method: the attribution is reliable only when a unique sparse support exists. On ESOL's named descriptors it recovers a Lipinski/Delaney-consistent nested set (polar surface area, molecular weight, rings) and is flagged reliable; on correlated bases (QM7 Coulomb eigenvalues, ArrowHead) no unique support exists and it is flagged *unreliable*. A controlled test confirmed the failure is correlation-driven, not dimension-driven: sixty independent features are recovered exactly, sixty correlated features are not.

*Tier 3, symbolic read-out.* A KAN [17, 13] radial's learned spline is projected onto an analytic dictionary by orthogonal matching pursuit with a residual-gated acceptance. This is the only tier with an intrinsic fidelity gap—the spline is ground truth and every closed form carries a residual—so a form is accepted only above an $R^2$ threshold and otherwise the spline is kept ("a fit, not the law"). In the direct/bottleneck regime it recovers $e^{-d}$, $e^{-d^2}$, $\frac{1}{2}d^2$ exactly and declines $\cos 2d$ and damped oscillators honestly; in the embedded regime (real trained equivariant nets, which distribute the physics across channels) it declines, correctly reporting that no single-channel closed form is faithful.

*Redundancy reduction* (`core/redundancy_reduction.py`) applies the participation-ratio functional one level down, to the covariance spectrum of the data, to measure an effective dimension $d_{\text{eff}}$ and project onto the leading $\lceil d_{\text{eff}} \rceil$ collective coordinates—the renormalization-group "keep the relevant modes" step, driven by measured spectral weight. On the QM7 Coulomb eigenvalues it measures $d_{\text{eff}} = 3.78$ of 23 and reduces to four dimensions while retaining $R^2$ 0.969 versus 0.979: an 83% dimension cut for a one-point accuracy loss. It is opt-in preprocessing that completes the kinematics rung—the symmetry group removes the exact redundancy, the effective dimension removes the approximate data-dependent redundancy that remains.

### 10.1 One ledger: effective dimension across the coarse-graining axis

The participation ratio that measures $d_{\text{eff}}$ above is not unique to the data covariance. The same functional $\mathrm{PR}(s) = (\sum_i s_i)^2/\sum_i s_i^2 = 1/\sum_i p_i^2$ (equal to 1 on a single-mode spectrum, $m$ on a uniform $m$-mode one) is what counts the effective number of primitives in the mixture ($\mathrm{IPR}(\boldsymbol{\alpha})$, §9.4), the two differing only in which spectrum it eats—the covariance eigenvalues in one case, the $\boldsymbol{\alpha}$ weights in the other. The supervised information-bottleneck flow $d_{\text{IB}}(\beta)$ resolves the same data-mode count against a renormalization-group scale, and the local learning coefficient $\lambda$ of §4.4 measures the effective dimension of the whole fitted *model*. These four grew up as separate measurements in separate modules; they are one question—how many degrees of freedom does this representation really use—asked at ascending levels of coarse-graining, and the package now states them on a single axis (`machinery/effective_dimension_ledger.py`, opt-in `report_ledger`).

The unification is exact where it is exact and bounded where it is not. Three of the four legs are *literally* the participation ratio—`participation_ratio` reproduces both the data $d_{\text{eff}}$ and the mixture IPR to machine precision, since it is the same $1/\sum p^2$—so the data-mode and primitive levels share a functional, not merely a metaphor. The fourth leg, $\lambda$, is a *different* functional (the RLCT, not a participation ratio) answering the same question at the model level, and is the natural cap of the ledger ($\lambda \leq k/2$). The honest boundary, stated in the ledger itself, is that the levels are not one commensurable scalar: a count of effective data modes and a count of effective primitives carry different units and are never summed. So the object is one functional read at the kinematic levels plus $\lambda$ at the model level—the quantitative completion of the package's "kinematics $\rightarrow$ degrees of freedom" ordering, stated once rather than as four vocabularies. On a representative fit it reads coherently across levels: $\sim$63 effective data coordinates of the ambient covariance, $\sim$4 effective primitives of the deployed mixture, and $\lambda$ far below $k/2$ (the singular signature of §4.4) at the model. Like the other read-outs it is pure reporting and changes no selection.

### 10.2 Developmental read-out: the complexity trajectory over training

The local learning coefficient of §4.4 is estimated *once*, at the converged optimum, because that is the only place the RLCT is meaningful. The developmental-interpretability programme [12] observes that the *trajectory* $\hat{\lambda}(t)$ along training carries additional structure: its plateaus and jumps mark staged learning—Bayesian phase transitions at which the model acquires new effective degrees of freedom. Metaopt's dynamics stage trains weights and reads out only at the end; `machinery/developmental_llc.py` (`developmental_llc`) adds the trajectory read-out, reusing the same estimator unchanged at a schedule of checkpoints. The content is the checkpoint schedule and the reading of the resulting curve, not a new estimator: after a fit the selected architecture is rebuilt as a fresh untrained net and trained once through a denser-early checkpoint schedule, with $\hat{\lambda}$ probed at each checkpoint and the curve read for the located transitions.

Its scope is stated honestly and was fixed by premise checks. The *convergence onset*—the sharp negative$\rightarrow$positive flip of $\hat{\lambda}$ as $w^\star$ settles into a genuine basin, coinciding with the training loss flattening—is robust and is the primary deliverable. Multi-stage sub-structure within the positive regime appears only when the data itself separates the learning phases (a small network with a single dominant mode shows one onset, not a staircase); it is therefore reported as advisory candidate jumps that exceed the per-checkpoint SGLD noise, never asserted. And the per-checkpoint values are a short SGLD estimate, so the deliverable is the *shape* and the located transitions, exactly as the developmental literature uses $\hat{\lambda}(t)$. The read-out is opt-in (`developmental_llc`), changes no selection, and leaves the deployed architecture and its metric provably unchanged.

### 10.3 Response spectroscopy: the curvature of the selection

The package prices every selection by an objective but reports only its arg min—never how *sharply* that minimum is preferred, nor how far it sits from the nearest selection boundary. The same devinterp line now frames precisely this as spectroscopy: infer internal structure by measuring the response to a perturbation [9]. `machinery/response_spectroscopy.py` (`report_response`) adds the second-derivative read-out, reusing quantities the fit already computed—the solo energies behind the Gibbs-$\alpha$ and the $\{\text{scores}, \Omega\}$ behind the contract bake-off—so it is a curvature measurement on an objective in hand, not a new optimisation. It is the natural second derivative of the thermodynamic potential of §9.3: it measures the curvature of the very Level-A and Level-C free energies that potential documents.

The premise checks decided its form, and the honest finding is that the two levels have *different* analytic character. At the primitive readout (Level A) the Gibbs measure is smooth, so the specific heat $\chi = \mathrm{Var}_\alpha(\Psi) = \partial^2(-\log Z)/\partial\beta^2$ is a genuine susceptibility—but it is non-monotone in $\beta$ (it peaks at the "melting" temperature and vanishes at both extremes), so the monotone decisiveness read is the readout entropy $H(\alpha)$, not $\chi$; both are reported, with the verdict keyed to entropy and the energy gap to the runner-up. At the contract choice (Level C) the priced objective $\mathcal{J}_c(\mu) = R_c + \mu\,\tilde{\Omega}_c$ is piecewise-*linear* in the price, so $\partial^2\mathcal{J}^\star/\partial\mu^2 = 0$ in the interior with the curvature concentrated at the winner-switch kinks: selection in $\mu$ is a sequence of *first-order* transitions, and the reported observable is accordingly the critical price $\mu^\star$ to the nearest switch, the price margins to flip, and the slope jump across the transition (the finite susceptibility of that first-order line), together with a robustness verdict calibrated on the natural scale of $\mu$. The read-out thus turns the point-estimate selection into a calibrated one—how sharply is this primitive preferred, how robust is this contract to the price—while, like the other tiers, changing nothing about what is selected.

# 11 Validation

This section consolidates the empirical evidence: controlled synthetic tasks with known optima (§11.1), real-data results across the eight contracts (§11.2), a matched-complexity baseline comparison (§11.3), a reading of the concrete architectures the selector deploys (§11.4), and the single dataset registry and runner framework that regenerates all of it (§11.5). Unless noted, numbers are regenerated from the current package via the runners of §12.

## 11.1 Synthetic validation

Three synthetic tasks with known ground-truth optima exercise the schema (depth 1, width 64, all eight primitives). Results are in Table 11, measured from the current package.

Two of these outcomes are worth emphasizing. The recall result (acc 1.000, $\boldsymbol{\alpha}_{\text{attention}} = 0.87$) is the cleanest demonstration in the package that the selector commits decisively to content-based routing when—and only when—the task requires it. The adding result is instructive in a different way: the presumed ground truth was a gated recurrent cell, but with attention available the schema discovered that the marked-value sum is a routing problem, not a memory problem, and selected `attention`. This is genuine cross-family search, not confirmation of a prior expectation.

## 11.2 Real-data validation

Five standard UCR/UEA time-series classification datasets, spanning sensor, motion-capture, image-outline, accelerometer, and appliance-power domains, exercise the schema on real data (depth 1, width 64, official train/test splits, $z$-normalized; long series average-pooled on the time axis to at most 120 steps so the per-timestep recurrent scan stays tractable). Table 12 reports test accuracy and the selected architecture.

All five datasets select a recurrent primitive, with the recurrent family dominant in every case. The one dataset with meaningful non-recurrent weight was ACSF1, where the fixed frequency-basis `spectral` primitive surfaced ($\boldsymbol{\alpha} \approx 0.16$) alongside the selected `gated` cell—consistent with the periodic structure of appliance-power signals. An honest caveat applies to the low weight on `conv` throughout: the `conv` core used in this sweep is a single causal 1-D convolution with a small kernel, far shallower than the deep multi-scale convolutional networks (ResNet, InceptionTime) that dominate the UCR archive in the literature. The finding is therefore "recurrence beats *our shallow* convolution at this scale," not a claim about convolution for time series in general. This caveat has since been partly addressed: a multi-scale dilated core `dilconv` (§8.2) was added to the sequence vocabulary and, on OSULeaf, is selected over the single-kernel `conv` and improves accuracy by 5.8 points—so on datasets with multi-scale temporal structure the conv-family is now competitive, exactly as the caveat anticipated.

### 11.3 Comparison with matched-complexity baselines

An accuracy in isolation is uninformative; the relevant reference is a fixed neural network of similar size trained identically. Table 13 compares the schema's selected architecture against three standard baselines—a plain gated recurrent unit (GRU), a two-layer 1-D convolutional network (CNN), and a single-hidden-layer multilayer perceptron (MLP)—at comparable parameter count, same preprocessing, same training budget.

At matched complexity the schema is competitive: it wins or ties the best fixed baseline on three of five datasets, trails narrowly on the other two, and never selects the worst-performing primitive family. The one clearly interpretable shortfall is ItalyPowerDemand, where a flatten-then-dense MLP (0.957, the smallest model at 1.7k parameters) beats every sequential model. This dataset is "effectively tabular"—its short 24-step series is really a feature vector—and the schema's mean-over-time readout cannot express the flatten-then-dense operation that is optimal there. This is a genuine expressiveness gap in the readout, not a selection error: given the realizations available, the schema chose reasonably, but the winning realization was absent from its menu.

We do not compare against full deep-learning state of the art (SOTA) (GunPoint 0.91–0.97, ItalyPowerDemand $\approx$ 0.955, OSULeaf $\approx$ 0.98 with deep multi-scale CNNs and heavy tuning). Those numbers use depth, multi-scale kernels, full-length series, and hyperparameter search that the single-layer, width-64, fixed-budget schema does not; the matched-complexity comparison is the fair one.

### 11.4 Selected network architectures

Because the schema discretizes to a single primitive per layer, the deployed network is that primitive's core plus a readout head—a concrete, ordinary neural network. Table 14 records what the metaoptimizer produced for each task: the primitive type, the (depth 1) layer structure, the hidden width, and the deployed parameter count.

All selections reported *in this section* are at depth 1 and fixed width 64. On the compact sequence datasets examined here, added depth did not improve the marginal-value objective—consistent with the priced-depth analysis, which predicts small $L^\star$ absent genuine compositional structure, and with four convergent negative results during development in which a second layer failed to beat a single layer on compact real data. Depth-1 is therefore the honest reported architecture for these datasets, not a fixed choice—and on the deeper relational contracts the metaoptimizer does select heterogeneous multi-layer architectures (§4.2). The width, by contrast, *was* fixed here; §7 lifts that restriction and reports the architectures obtained when width and depth are themselves metaoptimized.

### 11.5 The standard validation framework

The datasets accumulated across the project are consolidated into a single registry that both a quick (subset, smoke-budget) and a full (complete-data, benchmark-budget) runner draw from, so that every schema is exercised on at least one real natural-science or medical dataset. The registry spans seventeen datasets: physiological time series (ECG5000 cardiology, UCR botany/motion), medical imaging (BloodMNIST hematology, OrganMNIST3D radiology), molecular graphs (ESOL solubility, Tox21 toxicity, QM7/QM9 quantum chemistry), molecular dynamics (rMD17 with forces), and particle physics (JetNet jet tagging as unordered sets). Because classification and regression are mixed, results are reported on a consistent skill axis in $[0, 1]$: (accuracy − chance)/(1 − chance) for classification and $R^2$ for regression, both reading zero at the trivial baseline and one at perfect. Routed through the meta-router at a smoke budget, all datasets dispatched correctly and cleared their trivial baselines where the compute budget was adequate, landing within a few points of literature SOTA on the near-saturated tasks (e.g. ECG5000) and trailing—by an amount that tracks the reduced budget, not a routing error—on the most compute-hungry ones (3D convolution, molecular-dynamics energy). The full runner supports CUDA (NVIDIA GPU) and Apple-Silicon (MPS) devices for benchmark-scale runs. Both runners now expose and wire through the full set of switches developed above—`--select` (`argmax/gibbs/sparse`), `--sparsity_mu`, `--tensorize_mu`, and `--tiebreak`—so the whole priced ladder (derived-$\boldsymbol{\alpha}$ selection, sparsity-priced mixture, priced tensorization, and the contract MDL tie-break) is exercisable in one command.

### 11.6 End-to-end run with the full priced ladder enabled

Table 15 reports the quick suite with *every* priced mechanism active at once (`--select sparse --sparsity_mu 0.3 --tensorize_mu 0.05 --tiebreak`). The point is not benchmark accuracy—these are subset, smoke-budget runs—but that all thirteen datasets dispatch to a valid contract, select a sensible compact architecture, and clear their trivial baselines bar the two hardest regression targets, with all pricing simultaneously live.

Two behaviours are worth reading off the table. The sparsity price is visibly active: where one primitive suffices the mixture collapses (MNIST IPR 1.3 →`conv_dw`; Tox21 IPR 1.2 →`gcn`) and the deployed net is compact, whereas genuinely multi-primitive tasks keep a higher IPR (ECG5000 8.6). Auto-tensorization dispatched every grid correctly, and—verified separately—re-discovers the latent shape when a single-channel grid is flattened (MNIST 784 $\rightarrow$ $28 \times 28$); a bare-flattened multi-channel image is kept as a tensor, since flattening conflates the channel and spatial axes. The contract routing is now handled by the learned router (§4.2.1) rather than a fresh bake-off: seeded from full-budget outcomes, it sends the geometric molecular targets to the equivariant contract at inference cost, which is what recovers their skill relative to the reduced-budget set mis-routing. The $\Omega_{\text{struct}}$ charge still governs genuine near-ties within the bake-off when the router is unconfident and the budget is adequate; at a smoke budget the router's full-budget-seeded prediction is preferred over an under-converged bake-off, which is the correct behaviour and the reason the geometric targets no longer collapse to the cheap set contract.

## 12 Reproducibility

Every number in §11.1–§11.3 is regenerated by the runner scripts in `validation_runners/`, which ship with the package:

- `run_unified_synthetic.py`—the synthetic tasks (copy, adding, recall);
- `run_unified_ucr.py`—the schema on any aeon UCR/UEA dataset, with a `--bilevel` flag that holds out a validation split from train for the $\boldsymbol{\alpha}$ selection (the honest architecture-selection protocol) and a `--primitives` flag to restrict the vocabulary;
- `run_baselines_ucr.py`—the matched-complexity GRU/CNN/MLP baselines.

The architecture-size metaoptimization of §7 is reproduced by three further runners: `run_minimal_architecture.py` (select primitive, width, and depth at a given price), `run_frontier.py` (sweep $\mu$ to trace the fit–complexity frontier), and `run_penalized_ selection.py` (the compact-and-accurate differentiable penalty). All reported accuracies are on the official held-out test split; the model never trains on test data. One protocol caveat is recorded plainly: in the default runs the network weights and the architecture parameter $\boldsymbol{\alpha}$ are both trained on the train split, so while the test accuracy is legitimate, the architecture *choice* was not made on held-out data. The `--bilevel` flag exists precisely to make that choice defensible when required; it tends to produce flatter, less overfit $\boldsymbol{\alpha}$ distributions.

# 13 Status of the first report's open problems

The first report closed with a list of open problems. All have since been resolved and are documented in the body; they are collected here, each stated in brief with a pointer and its honest residual limit, alongside two recently added contracts and the two predictions that were confirmed *negative*. The detail lives in the referenced sections.

## 13.1 Resolved since the first report

**Representation and size, unified under one price.** The *contract* choice, the *tensorization* choice, and the tensor *shape* are all now folded into the same action $J = R + \mu\,\Omega$ that prices primitive, width, and depth (§4.3, §5.3), with significance gates that make them safe to run by default. "Which interface" and "which tensor shape" are thus metaoptimality decisions rather than hand-set charges. The one-time $O(D^2)$ mutual-information cost is handled by an on-demand sampled estimator that never forms the full matrix. *Residual:* automatic 3D tensorization still reports only the leading 2D stride, so a flat vector is not yet auto-routed to the volumetric contract (parsing multiple stride peaks into a full $(D, H, W)$ shape is the remaining step); and two fail-safe signal-strength limits (background-dominated or exotic-aspect-ratio volumes) may under-rank, never mis-rank.

**Nonlinear symmetries.** The linear symmetry front-end (§6) is now backed by a LaLiGAN-style symmetry-regularized autoencoder that linearizes a nonlinear group action, after which the linear detector runs in the latent space; a null-baseline guard (the candidate generator must be decisively quieter than random generators of matched size) suppresses the false positives the joint objective is otherwise prone to, and a linear-first escalation tries the cheap exact detector before falling back. *Residual:* recovery is bounded by the autoencoder's ability to linearize—moderate warps of a rotation are recovered, a severe coordinate warp still defeats it.

**Robustness of discrete detection.** Beyond the `coordinate_structure` guard, a gradient-sensitivity-weighted equivariance error (a selectable `error_mode`) normalizes the transformation error by the root-mean-square (RMS) gradient along the tested direction, so a coordinate the model barely depends on cannot contribute a spurious symmetry (§6.6). It costs one extra backward pass and, in head-to-head tests, recovers a genuine swap symmetry the scale-aware error misses without false-positiving on a near-flat direction.

**Expressiveness gaps and the performance bottleneck.** The ItalyPowerDemand shortfall (§11.3) is addressed by a held-out bake-off between the mean and a flatten-then-dense readout (charged its extra cost, guarded to short sequences, off by default), which selects flatten on the tabular-like case (validation $0.75 \to 0.96$) and leaves longer series on the mean; the `conv` gap was closed by the multi-scale `dilconv/atrous` cores (§8.2). The recurrent per-timestep Python loop that dominated wall-clock time is removed for the linear `linssm` primitive by an associative block-cumsum parallel scan—numerically identical to the loop in output and gradients, $\sim$1.6–1.7× faster on forward. *Residual:* the input-dependent `selssm` is not associative and keeps its loop; on CPU the full forward+backward is only break-even at the safe block size, so the larger training speedup awaits GPU or a log-space Blelloch scan.

**Depth and heterogeneous architectures.** The question of whether the schema produces genuinely *heterogeneous* per-layer architectures (all first-report selections were depth 1) is settled affirmatively (§4.2): on the deeper relational contracts the metaoptimizer selects distinct primitives at distinct layers (ESOL `gin`→`pna`→`pna`; rMD17 `e_painn`→`e_painn`→`e_tp`). The $L^\star = 1$ observation for *compact sequence* datasets stands as a property of those datasets, not the machinery.

**Folding symmetry and width into the size objective (investigated, honest boundary).** The invariant-versus-unconstrained choice is *already* a priced $\boldsymbol{\alpha}$-selection (§6.5), so a separate penalty is cosmetic. For width, three differentiable realizations (sigmoid gate, $L_0$ hard-concrete, differentiable-network-channel-pruning (DNCP) softmax over widths) were prototyped and fail to reproduce the sweep's elbow: channel-level pruning measures channel *redundancy* within one over-wide network, a different quantity from the whole-network *capacity saturation* the sweep measures. A differentiable width matching the sweep would need decoupled per-width backbones—no cheaper than the sweep. The sweep's *cost*, not its correctness, is the target, and the orthogonal parallel-scan route above is the right lever; the priced sweep (with its epoch-floor and uninformative-sweep guards) is retained.

**The standing theoretical seam.** The architecture parameter $\boldsymbol{\alpha}$, flagged in the first report as a bolt-on, is now *derived* (§9.2): the free energy $\mathcal{F}_{\boldsymbol{\alpha}} = \langle \boldsymbol{\alpha}, \Psi \rangle - \frac{1}{\beta} H(\boldsymbol{\alpha})$ on the primitive simplex makes $\boldsymbol{\alpha}$ the outer Gibbs measure $\boldsymbol{\alpha}_p \propto e^{-\beta \Psi_p}$, its Fisher–Rao gradient flow the replicator equation, and DARTS that same flow in a mildly preconditioned metric. The variational principle *forces* the energy to be the *solo* marginal risk, so the $\boldsymbol{\alpha}$-derivation and the co-adaptation fix (§9.1) are one result. *Residual:* the cross-contract dispatch is necessarily discrete (§4.2), so the unification is complete *within* a contract but stops, by the interface-incompatibility argument, at the contract boundary.

**Bilevel versus joint selection, across datasets.** The first report flagged that the honest $\boldsymbol{\alpha}$-selection protocol (weights trained on one split, $\boldsymbol{\alpha}$ selected on a disjoint held-out split) was implemented but not the default in the reported sweeps, and asked for a systematic comparison against joint selection ($\boldsymbol{\alpha}$ and weights trained together) to check that the selections are defensible rather than capacity-chasing artifacts. This was run (seed-averaged mixture weights as the selection signal, since a single argmax is seed-dominated at these scales; test accuracy disjoint from both splits). The finding is nuanced and honest. Bilevel does what the theory predicts—wherever joint places bimodal or heavy mass on the high-capacity `lstm`, bilevel pulls it toward lower-capacity, more appropriate primitives and its models never memorize (train accuracy stays well below one), direct evidence the selections are not mere capacity artifacts—and, when there is enough data to split, matches joint's mean accuracy at *lower* seed variance. But the split *halves* the training pool, and on small datasets that cost dominates: on a 67-example task bilevel was far worse (test

$0.77 \pm 0.14$ versus joint $0.91 \pm 0.03$), while on a task with a genuine `lstm` trap it slightly improved the mean. *Resolved as a boundary, not a default:* joint selection on the full pool is the right default (the priced/clean-solo guards of §9.1 already curb capacity-chasing without paying a data cost), and bilevel is the right tool specifically when the pool is large enough to spare half and the vocabulary contains a high-capacity primitive that could memorize (see `studies/bilevel_vs_joint_study.py`).

### 13.2 Two recently added contracts

**Learnable-radial (KAN) equivariant primitives.** Two learnable-spline (Kolmogorov–Arnold network, KAN) siblings were added to the equivariant contract: `e_kan` and `e_painn_kan` replace the fixed Gaussian-basis radial of the tensor-product and PaiNN-style primitives with a learnable univariate spline (the non-radial primitives correctly acquire no variant). Both are exactly equivariant. The learned radial is directly inspectable—it recovers a Lennard-Jones curve to $R^2 \approx 0.96$ and beats the fixed basis on a sharp double-well (0.84 vs 0.63)—while tying on smooth electrostatic targets, and the metaoptimizer selects between fixed- and spline-radial per layer (choosing `e_painn_kan` in the end-to-end QM7 selection). The inspectable radial is a concrete bridge to interpretability.

**The neural-operator contract.** An eighth contract handles function-valued (partial-differential-equation, PDE-like) data—a map from $a(x)$ to $u(x)$ whose defining property is discretization invariance (weights live in Fourier-mode space, so training at one resolution evaluates at another), distinct from the `spectral` sequence primitive that collapses the spectrum to statistics. It carries its own $\boldsymbol{\alpha}$-simplex over three operator-layer families—spectral (Fourier neural operator at two mode budgets), local (Nemytskii), and global-low-rank (DeepONet-style [18] branch–trunk)—and is dimension-generic via the $n$-dimensional real fast Fourier transform (FFT). It validates end to end on 1D Burgers ($R^2 \approx 0.999$, zero-shot to higher resolution), 2D Darcy ($\approx 0.96$), and 3D heat diffusion ($\approx 0.997$). The truly *mesh-free* DeepONet (evaluable at arbitrary off-grid points, which the FFT-bound layers cannot be) is provided as a standalone architecture outside the grid-locked pipeline and validates on the antiderivative benchmark ($R^2 \approx 0.99$ on-grid, $\approx 0.96$ off-grid).

### 13.3 Confirmed negatives

**Confirmed negative results, restated.** Two of the theory's sharper predictions did not survive contact with data, and these are results, not gaps. The Route-1 depth-freedom prediction—that at criticality the flow's arc length (its complexity) stays bounded for ReLU ($\alpha = 2$) while growing only logarithmically for tanh ($\alpha = 1$)—was refuted directly (`tests/crit_arclen.py`). Trained flows do not stay critical (the leading Jacobian eigenvalue real-part settles well away from zero even under a criticality regularizer), and once off criticality the arc length grows at least linearly and often explosively in the integration time for *both* activations, with ReLU *less* bounded than tanh— the opposite of the predicted ordering. The exponent story is a critical-point statement that the trained network simply does not occupy.

**No compact real task where an advanced primitive is canonically optimal.** The search for a compact real dataset on which an "advanced" primitive (LSTM or attention) is *canonically* optimal repeatedly failed the same way: compact real tasks are dominated by local or simple structure that convolution and dense capture data-efficiently, so the advanced primitives' genuine advantages appear only on synthetic structured tasks (long-range memory for gating/LSTM, content-addressed routing for attention). The bilevel study above is consistent with this: where joint selection did place heavy mass on `lstm` for a compact real series (GunPoint), that mass was a capacity-chasing artifact that the held-out protocol pulled back, not a genuine optimum. The metaoptimizer reflects this honestly—it selects the advanced primitive exactly when the task's structure rewards it, and was never observed to select a worse-performing primitive.

# 14 Provenance and references

The package is an integrative repackaging of published research: each primitive and each selection rule realizes an idea from the literature, recombined under the common minimum-description-length action $J = R + \mu\Omega$ and the softmax primitive relaxation. Citations on first use above point to the originating works; the consolidated list follows. The contribution here is the unification across contracts and the honest empirical auditing of when each borrowed component pays for itself—not the components themselves.

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| **Analytical formulation** | **Status** | **Numerical realization** |
|---|---|---|
| Weight optimization (inner saddle: forward + backprop) | Exact | Standard autograd + Adam; unchanged from ordinary training. |
| Width as atomic measure; total-variation (TV)/atomic-norm sparsity with finite-atom certificate | Exact (2-layer) | Dual-certificate and column-generation solvers in `machinery/`; the certificate stopping rule terminates at a finite, $\lambda$-controlled neuron count. |
| Wasserstein / mean-field training dynamics | Approximate | Realized implicitly through gradient descent on the finite-width network; the measure flow is not integrated explicitly. |
| Priced depth (Route 2), $L^\star \sim \mu^{-1/(\alpha+1)}$ | Approximate | Marginal-value stopping rule over integer depth, now wired into the schema pipeline (§7); the closed-form exponent scaling remains heuristic. |
| Geometric depth (Route 1, neural-ODE geodesic) | Approximate | `neural_ode.py` integrates a continuous-depth flow; the depth-freedom prediction was tested and *not* confirmed (constant-speed flow). |
| Depth renormalization-group (RG) criticality; critical $\sigma_w^2$ | Exact (diagnostic) | `core/meanfield.py` locates $\chi_1 = 1$ and classifies phases; used to initialize recurrent cores. |
| Architecture parameter $\boldsymbol{\alpha}$ (primitive selection) | Derived | Differentiable-architecture-search (DARTS [16]) style softmax mixture, now derived as the outer Gibbs measure $\boldsymbol{\alpha}_p \propto e^{-\beta\Psi_p}$ on the primitive simplex (§2.3, §9.2); DARTS is its Fisher–Rao gradient flow. |
| Six-primitive irreducible vocabulary | Realized | Eleven selectable sequence *compositions* of the six irreducibles, plus per-contract realizations (§8.1). |
| Architecture-crossing / class-invariance | Partial | Realized only within the fixed primitive vocabulary; inventing a primitive outside it remains out of scope by construction. |

Table 1: Mapping from analytical formulations to their numerical status in the package.

| Contract | Interface / group | Selectable primitives |
|---|---|---|
| sequence | $(b, T, c)$; time-shift | plain, gated, lstm, conv, **dilconv**, attention, dense, norm, spectral, linssm, selssm (11) |
| spatial | $(b, c, H, W)$; 2D translation | conv2d, conv2d_k5, conv2d_k7, **atrous**, conv_dw, pointwise, dense, norm, attention (9) |
| volumetric | $(b, c, D, H, W)$; 3D translation | conv3d, conv3d_k5, conv_dw, pointwise, dense, norm, attention (7) |
| 4D | $(b, c, T, D, H, W)$; 4D translation | conv4d, conv4d_kt1, conv_dw, pointwise, norm (5) |
| graph | nodes + edges; Aut$(G)$ | gcn, sage, gin, **pna**, gat, dense, norm (7) |
| equivariant | nodes + positions; SO(3) | e_tp, **e_kan**, **e_painn**, **e_painn_kan**, e_mean, e_linear, e_gate, e_norm (8) |
| set | unordered elements; $S_n$ | deepsets, sab, isab, pma_block, element_mlp, norm (6); poolings sum/mean/max/pma |
| operator | function on a grid; discretization-invariance | fourier, fourier_wide, local, deeponet (4) |

Table 2: The eight schemas. Each realizes the six irreducibles (affine, pointwise, gating, weight-sharing-under-$G$, content routing, normalization) under its own group and tensor rank. Cores added after the initial report are set in bold (the multi-scale **dilconv/atrous** of §8.2, the graph aggregator **pna**, and the learnable-radial **e_kan/e_painn_kan** of §13). The set contract is the maximal-symmetry endpoint ($S_n$); the grid ladder sequence→spatial→volumetric→4D is a tower of translation groups of increasing rank, each a hard structural boundary because the $N$-D translation group is not a subgroup of the $(N-1)$-D one; the operator contract sits apart, mapping a sampled function to a function with weights in Fourier-mode space so that it is discretization-invariant.

| Target | Fit ($R^2$): set / graph / equiv | Selected | Why |
|---|---|---|---|
| geometric (radius of gyration) | −0.07 / −0.09 / **0.67** | equivariant | fit decides; charge irrelevant |
| topological (mean degree) | −0.05 / **0.956** / 0.847 | graph | graph fits best; charge cements |
| set (mean feature) | **0.948** / 0.65 / 0.944 | set | near-tie set≈equiv; $\Omega_{\text{struct}}$ picks cheaper |

Table 3: The derived contract charge on controlled targets ($\mu_c$=0.05; $\Omega_{\text{struct}}$: set 0, graph ∼10, equiv ∼58 nats). The *set* row is the decisive case: a genuine fit near-tie (0.948 vs 0.944) resolved entirely by the derived structural code length toward the cheaper set contract—exactly what the ordinal $\kappa$ did by hand, now principled. The charge bites only at near-ties; when one contract fits best (geometric), fit alone decides.

| Dataset (true shape) | Parsed | Ratio | Verdict |
|---|---|---|---|
| MNIST (28×28) | 28×28 | 5.8 | correct |
| BloodMNIST (28×28) | 28×28 | 4.9 | correct |
| OrganMNIST3D ($4^3$, $8^3$ crop) | $4^3$, $8^3$ | $2.6, 1.9$ | correct |
| OrganMNIST3D ($7^3$ strided) | 1D | 4.5 | under-ranks (fail-safe) |
| ECG5000, GunPoint, Italy-Power | 1D | – | correct |
| Unstructured noise | 1D | 1.1 | correct (gate) |

Table 4: Priced tensorization on real data (ratio = neighbour MI over off-diagonal baseline; the significance gate promotes beyond 1D only above ∼1.6). Synthetic reliability, over five seeds per shape, is 100% for cubic/near-cubic lattices, 2D grids and sequences (including rank-4 4×4×4×4), degrading only on fully-distinct-axis non-cubic shapes, which are refused rather than mis-ranked.

| Dataset | Price $\mu$ | $K^\star$ | $L^\star$ | Primitive(s) | Test acc |
|---|---|---|---|---|---|
| ItalyPowerDemand | 0.001 | 16 | 2 | `lstm`, `spectral` | 0.819 |
| | 0.008 | 8 | 2 | `plain`, `conv` | 0.603 |
| GunPoint | 0.001 | 16 | 1 | `attention` | 0.687 |
| | 0.008 | 8 | 2 | `conv`, `attention` | 0.493 |
| | 0.020 | 8 | 1 | `attention` | 0.493 |

Table 5: Fit–complexity frontier: metaoptimized width $K^\star$, depth $L^\star$, and primitive(s) as the capacity price $\mu$ varies. The selected architecture—size, depth, and primitive—shifts along the frontier and differs between datasets: ItalyPowerDemand justifies a second layer, GunPoint mostly does not.

| Dataset | $K^\star$ | Primitive | Params | Test acc | Fixed-64 ref |
|---|---|---|---|---|---|
| ItalyPowerDemand | 64 | `spectral` | 44,301 | 0.813 | 0.928 |
| GunPoint | 8 | `linssm` | 4,261 | 0.653 | 0.893 |
| BasicMotions | 32 | `linssm` | 17,625 | 0.975 | 0.975 |
| OSULeaf | 32 | `spectral` | 14,641 | 0.393 | 0.607 |
| ACSF1 | 32 | `plain` | 14,773 | 0.330 | 0.400 |

Table 6: Compact-and-accurate selection via the differentiable complexity penalty ($\mu = 0.3$, accuracy-first width compaction) over the nine-primitive vocabulary (including `linssm`). "Fixed-64 ref" is the joint-train accuracy at fixed width 64 (Table 12). The method compacts where validation accuracy permits (GunPoint to 8, BasicMotions and OSULeaf and ACSF1 to 32) and holds width where it is needed (ItalyPowerDemand). Notably `linssm` is selected on BasicMotions (test 0.975 at width 32, half the fixed-64 width) and surfaces as the argmax on GunPoint—the new linear-recurrence primitive earns selection under the honest chosen-size protocol, not merely at fixed width. Accuracies run below the fixed-64 joint-train references because compaction and the bilevel (held-out-validation) protocol both cost some accuracy; the tradeoff, not a regression, is the object of interest.

| Code entry | Irreducibles used | Note |
|---|---|---|
| `dense` | affine + pointwise | The least-biased value transform. |
| `plain` | affine + pointwise + sharing(time) | Plain recurrence. |
| `conv` | affine + pointwise + sharing(space) | Same primitive (4), different group. |
| `dilconv` | affine + pointwise + sharing(space), multi-scale | Dilated/multi-scale conv; a `conv` variant with an exponentially growing receptive field. |
| `gated` | + gating | GRU-style multiplicative interpolation. |
| `lstm` | + gating | A *different composition* of the same irreducibles (additive cell carry). |
| `attention` | routing | The one-to-one realization of primitive (5). |
| `norm` | normalization + affine | The depth stabilizer. |
| `spectral` | (non-irreducible) | Fixed frequency-basis primitive; conjugate to `conv`, not one of the six. |
| `linssm` | affine + sharing(time), linear | Diagonal linear state-space recurrence (S4 [11]/S5 [22]/Mamba [10] core); a non-irreducible recurrence variant with explicit decay timescale. |
| `selssm` | affine + gating + sharing(time) | Input-selective (Mamba-style) state-space recurrence; the gated, input-dependent sibling of `linssm`. |

Table 7: The eleven selectable sequence entries as compositions of the six irreducible primitives. Pointwise nonlinearity (2) never appears standalone—it is folded into every core as the activation. Weight sharing (4) appears twice (as `plain` and `conv`) because the group does not relax smoothly and must be offered as separate slots. Gating gives two nonlinear entries (`gated`, `lstm`) that share irreducibles but differ in composition and optimization behavior. `spectral`, `dilconv`, `linssm`, and `selssm` are useful non-irreducible additions—`dilconv` a multi-scale convolution, `linssm` a linear (associative, parallelizable-in-principle) recurrence, and `selssm` its input-selective nonlinear sibling.

| Module | Interface | Primitive set / role |
|---|---|---|
| `supergraph.py` | recurrent | 2-way {plain, gated}; first selection machinery (validated). |
| `multi_supergraph.py` | recurrent | $N$-way {plain, gated, lstm}; adds LSTM and chrono-init. |
| `parallel_supergraph.py` | flat vector | 2-way {conv, spectral}; conjugate-basis antichain (validated). |
| `multi_parallel_supergraph.py` | flat vector | $N$-way {conv, spectral, attention, dense, norm}. |
| `spatial_supergraph.py` | 2-D image | 2-way {conv, dense}; spatial selection. |
| `schema.py` | sequence | 8-way, all realizations; the capstone (§8.3.1). |

Table 8: The six schema modules. The first five split the primitives by input interface (recurrent, flat, spatial); the schema merges them onto a common sequence interface so that all primitives compete on one task.

| Dataset | argmax (mixture) | Gibbs (deployed solo) | params |
|---|---|---|---|
| ESOL | 0.655 `pna,pna,norm` | **0.663** `pna,pna,pna` | 127k→ 91k |
| Tox21 | 0.621 `gin,norm,norm` | 0.601 `gcn,gcn,gcn` | 127k→ **7.9**k |
| QM7-graph | **0.665** `gin,norm,norm` | 0.504 `gin,gin,gin` | 126k→ 14k |
| rMD17 | **0.114** `e_painn,e_painn,e_tp` | 0.063 `e_tp`×3 | 20k→ **6.0**k |
| QM7-equiv | **0.759** `e_tp,e_norm,e_tp` | 0.718 `e_tp`×3 | 20k→ **6.0**k |

Table 9: Deployed derived selection versus the argmax mixture (held-out skill; params before/after). In every case the derived selection removes the parameter-free co-adapter (`norm`/`e_norm`) tail the argmax leaves, and deploys a 3–16× smaller network. On ESOL it also beats the mixture; where it trails (most on QM7-graph) the mixture's *combined* capacity genuinely helps—the honest distinction between *selection* (which primitive is best alone, which the derivation answers correctly and robustly) and *deployment* (whether a mixture beats any one primitive, sometimes yes). The same $\beta$ interpolates the two: finite $\beta$ retains the mixture, $\beta \to \infty$ commits.

| Dataset | argmax (dense) | sparse $\mu$=0.3 | sparse $\mu$=0.6 |
|---|---|---|---|
| ESOL | 0.655, IPR∼5 | 0.599, IPR 5.0 | **0.748**, IPR 5.0 |
| Tox21 | 0.621, IPR∼5 | 0.660, IPR 1.23 | **0.693**, IPR **1.09** |
| QM7-graph | 0.665, IPR∼5 | 0.658, IPR 4.9 | 0.662, IPR 4.5 |

Table 10: Sparsity-priced $\boldsymbol{\alpha}$ on the graph contract (held-out skill; IPR = effective number of primitives, Eq. 2). *Tox21* collapses to a compact ∼1-primitive mixture and *improves* (the price cuts a co-adaptation-inflated dense mixture down to a cleaner model that generalises better). *QM7-graph* stays dense and roughly flat—it genuinely wants its mixture and the frontier correctly declines to collapse it. *ESOL* improves at $\mu$=0.6 while staying dense: honestly, that gain is the price acting as a training *regulariser* (the $-\sum \boldsymbol{\alpha}^2$ term perturbs the $\boldsymbol{\alpha}$ trajectory), not compression—reported as such. The point is the per-dataset spread from one knob; a fair deployment selects $\mu$ per dataset by the elbow/validation rule (`select_sparsity_by_elbow`) rather than sharing it.

| Task | Structure | Performance | Selected (peak $\boldsymbol{\alpha}$) |
|---|---|---|---|
| Copy (delay 15) | recurrent memory | acc 0.995 | `plain` (0.48; recurrent family ≈ 0.86) |
| Adding ($T$=80) | long-range sum | mse 0.0008 | `attention` (0.26) |
| Assoc. recall ($N$=8) | content routing | acc 1.000 | `attention` (0.87) |

Table 11: Schema on synthetic tasks. Copy routes to the recurrent family (`plain` suffices at delay 15); recall selects `attention` decisively; adding selects `attention`, which routes to the two marked positions and sums them—a better solution than carrying a running sum through a recurrent scan, and one the metaoptimizer found by searching the full vocabulary rather than a recurrent-only subset.

| Dataset | Domain | Test acc | Major. | Classes | Selected |
|---|---|---|---|---|---|
| ItalyPowerDemand | sensor/power | 0.928 | 0.501 | 2 | `plain` |
| GunPoint | motion capture | 0.893 | 0.507 | 2 | `lstm` |
| OSULeaf | image outline | 0.607 | 0.227 | 6 | `gated` |
| BasicMotions | accel. (6-ch) | 0.975 | 0.250 | 4 | `lstm` |
| ACSF1 | appliance power | 0.400 | 0.100 | 10 | `gated` |

Table 12: Schema on five real UCR/UEA datasets. Every accuracy is well above the majority-class baseline. All five select a recurrent primitive: the schema reads time-series classification as fundamentally temporal at this scale. `spectral` carried notable weight (≈ 0.16) only on ACSF1, where appliance-power signals have genuine periodic structure.

| **Dataset** | **Ours (sel.)** | **GRU** | **CNN** | **MLP** | **Ours vs best** |
|---|---|---|---|---|---|
| ItalyPowerDemand | 0.928 (`plain`) | 0.539 | 0.549 | **0.957** | −0.029 |
| GunPoint | **0.893** (`lstm`) | 0.493 | 0.660 | 0.773 | +0.120 |
| OSULeaf | **0.607** (`gated`) | 0.467 | 0.397 | 0.492 | +0.115 |
| BasicMotions | 0.975 (`lstm`) | 0.750 | **1.000** | 0.900 | −0.025 |
| ACSF1 | 0.400 (`gated`) | 0.280 | **0.400** | 0.180 | ±0.000 |

Table 13: Schema vs matched-complexity fixed baselines (test accuracy). The schema wins outright on two datasets, ties the best on one, and trails narrowly on two. It consistently avoids the *worst* baseline—on GunPoint a plain GRU collapses to near chance (0.493) while the schema's `lstm` selection reaches 0.893.

| **Task / dataset** | **Primitive** | **Layers** | **Width** | **Deployed params** |
|---|---|---|---|---|
| Copy | `plain` | 1 | 64 | 5,062 |
| Adding | `gated` | 1 | 64 | 12,929 |
| Assoc. recall | `attention` | 1 | 64 | 6,054 |
| ItalyPowerDemand | `plain` | 1 | 64 | 4,354 |
| GunPoint | `lstm` | 1 | 64 | 17,026 |
| BasicMotions | `lstm` | 1 | 64 | 18,436 |
| OSULeaf | `gated` | 1 | 64 | 13,062 |
| ACSF1 | `gated` | 1 | 64 | 13,322 |

Table 14: The concrete network selected per task (primitive core + linear readout head, width 64, depth 1). Deployed parameter counts vary with the primitive's gate multiplicity (`plain` carries one weight matrix per state update, `gated` three, `lstm` four) and with the input/output dimensions of the task. The full depth-1 schema from which each is selected is larger—41k to 71k parameters depending on input dimension—since it instantiates all eight primitives in parallel; discretization to the selected core is what yields the compact deployed network.

| **Dataset** | **Contract** | **Metric** | **Skill** | **Architecture** | **IPR** | **Params** |
|---|---|---|---|---|---|---|
| ECG5000 | sequence | 0.917 | +0.90 | lstm | 8.6 | 14,672 |
| GunPoint | sequence | 0.493 | −0.01 | lstm | 9.0 | 14,573 |
| OSULeaf | sequence | 0.306 | +0.17 | plain | 9.0 | 14,705 |
| BloodMNIST | spatial | 0.735 | +0.70 | atrous | 4.4 | 6,573 |
| MNIST | spatial | 0.760 | +0.73 | conv_dw | 1.3 | 6,319 |
| OrganMNIST3D | volumetric | 0.289 | +0.22 | conv_dw | 4.0 | 3,785 |
| ESOL | graph | 0.599 | +0.60 | pna→gin→norm | 5.0 | 126,790 |
| Tox21 | graph | 0.660 | +0.66 | gcn→gcn→norm | 1.2 | 126,839 |
| QM7-graph | graph | 0.658 | +0.66 | gin→norm→ norm | 4.9 | 126,358 |
| QM7-equiv | equiv* | 0.768 | +0.77 | e_tp→e_norm→ e_tp | 3.9 | 20,449 |
| QM7-set | set | 0.579 | +0.58 | norm→norm | 3.0 | 29,335 |
| JetNet | set | 0.357 | +0.20 | deepsets→ element_mlp | 1.8 | 29,243 |
| rMD17-ethanol | equiv* | 0.151 | +0.15 | e_painn→ e_painn→e_tp | 3.6 | 20,401 |

Table 15: Quick suite with the full priced ladder enabled (sparse selection, priced tensorization, contract tie-break). Skill is (acc − chance)/(1 − chance) (classification), $R^2$ (regression), or receiver-operating-characteristic area under the curve (ROC-AUC) (Tox21). IPR is the effective number of primitives in the deployed mixture (a low IPR means the sparsity price collapsed the mixture to essentially one primitive). "Params" is the *deployed* network, 3.7k–127k—one to two orders of magnitude below the deep, heavily-tuned SOTA references (ResNet/InceptionTime, message-passing neural network (MPNN)/SchNet). *QM7-equiv and rMD17 route to the *equivariant* contract via the learned router (§4.2.1), seeded from full-budget bake-off outcomes; this corrects the reduced-budget mis-routing to set (where the $\Omega_{\text{struct}}$ charge over-penalises the under-converged equivariant branch) and lifts skill substantially—QM7-equiv $R^2$ 0.57 → 0.77 and rMD17 $R^2$ −0.00 → 0.15, the latter from below the trivial baseline to genuinely predictive SO(3)-equivariant message passing. Disabling the router (`--no_learned_router`) reproduces the set mis-routing. The strongest at this budget are ECG5000 (0.917 at 15k params, versus 0.94–0.95 SOTA), QM7-equiv, and the graph tasks.